

# A three-body waltz: numerical search for recurring dynamics in the three-body problem

PhD defence – Alexandre Prieur – 23/09/2026

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## Jury

- Àlex Haro (**Reviewer**)
- Elisa Maria Alessi (**Reviewer**)
- Éric Gourgoulhon (**Examiner**)
- Gabriella Pinzari (**Examiner**)
- Jean-Baptiste Caillau (**Examiner**)
- Jérôme Daquin (**Examiner**)
- Philippe Robutel (**Supervisor**)
- Jacques Fejoz (**Supervisor**)



# Introduction

*And then there were three*



# The dread of an empty universe

In my mind, the first “emptiness” I created was the **infinity of space**. There was nothing in it, not even light.

— *The Three-Body Problem*, Liu Cixin, translated by Ken Liu, 2014

# The first sphere

So I created a **sphere** in this infinite space for myself: not too big, though **possessing mass**. [...] It hung there, never moving, never changing, like a perfect interpretation for death.

# The first sphere

So I created  
possessing m  
perfect interp



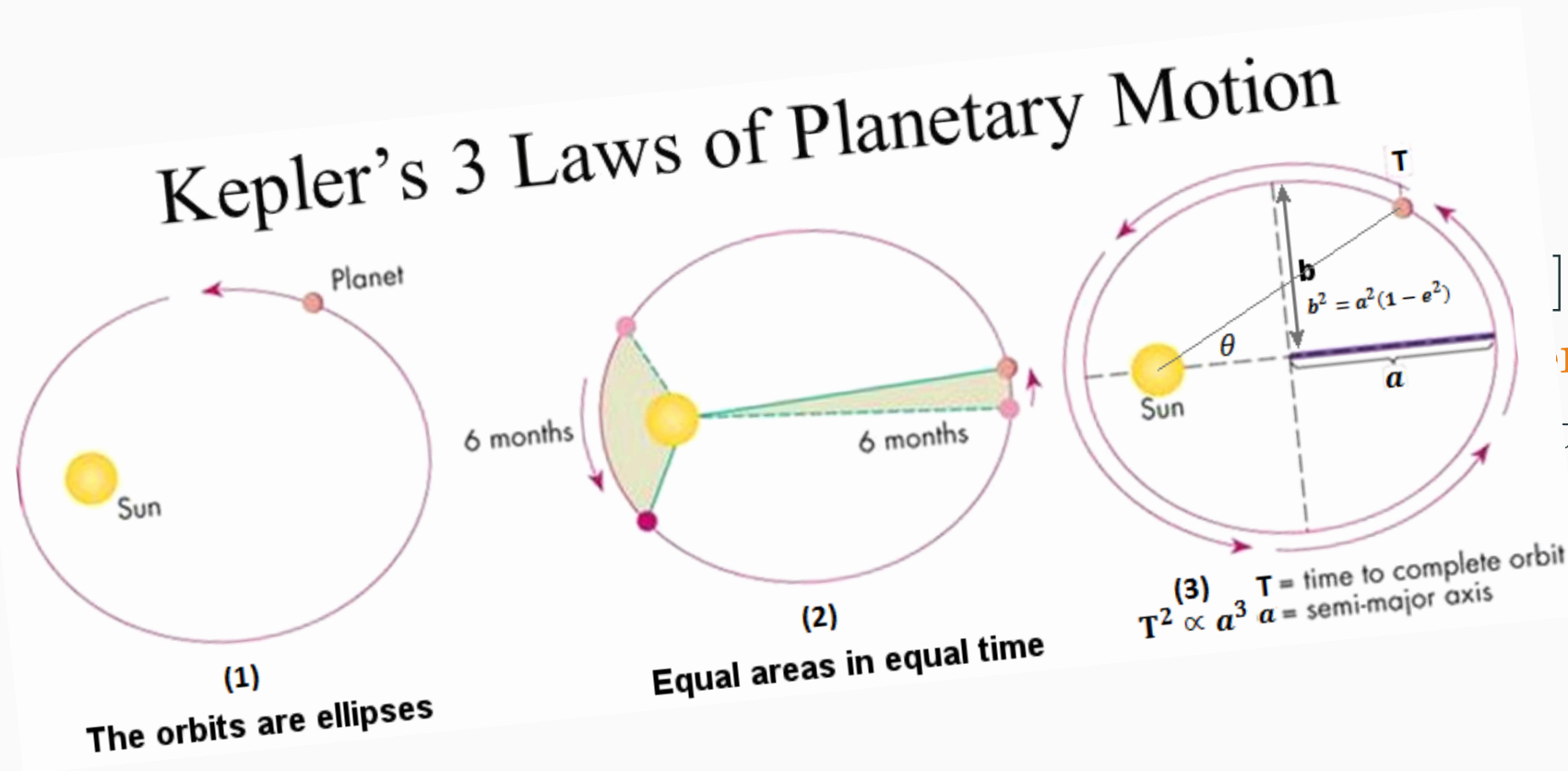
too big, though  
hanging, like a

# The second sphere

I created a **second sphere** whose mass was equal to the first one's. [...] they would be quickly pulled together by their own **gravitational attraction**. [...] No matter what the initial conditions, the revolutions would eventually stabilize and become unchanging: the **dance of death**.

# The second sphere

I crea  
would  
No m  
stabili

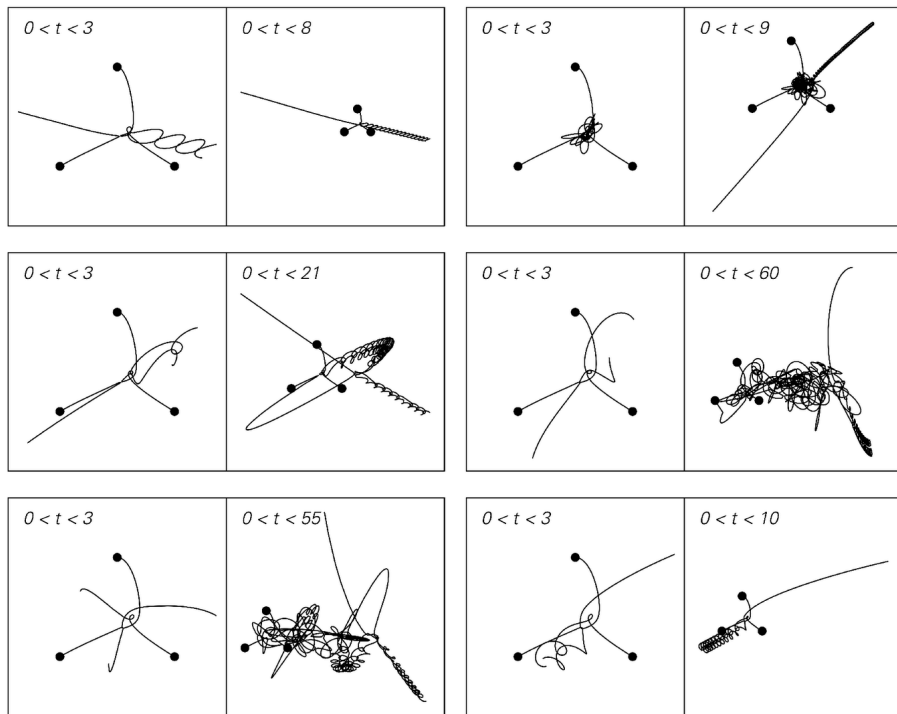


] they  
n. [...]  
ually

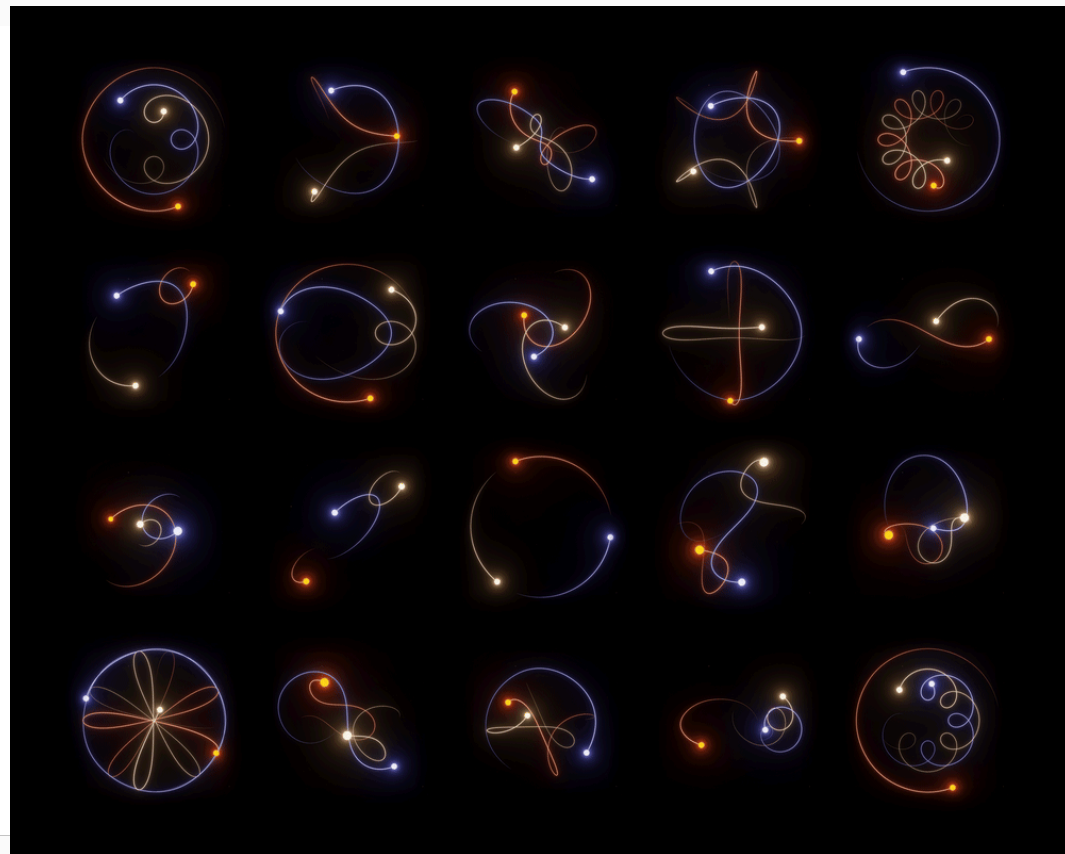
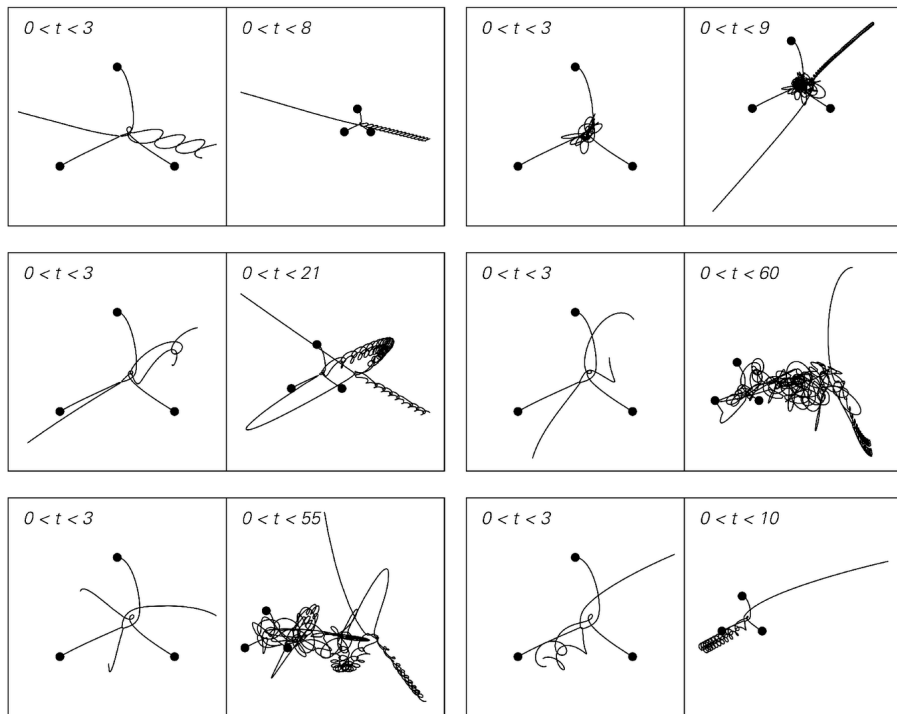
# The third sphere

I then introduced a **third sphere**, and to my astonishment, the situation changed completely. [...] The sphereless, one-sphere, and two-sphere universes all showed up as a single equation or a few equations, like a few lonesome leaves in late fall. But this third sphere gave “emptiness” life. **The three spheres, given initial movements, went through complex, seemingly never-repeating movements.** The descriptive equations rained down in a thunderstorm without end.

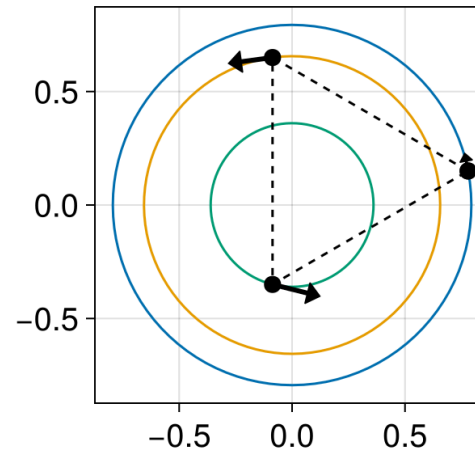
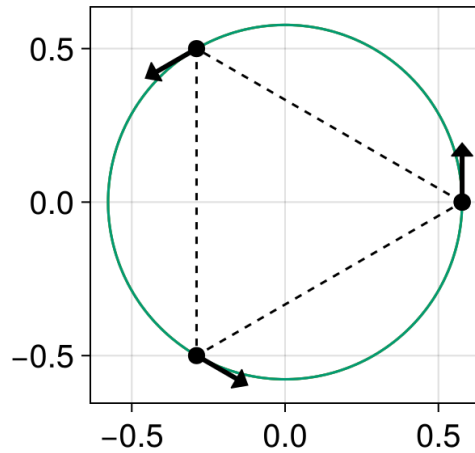
# The third sphere



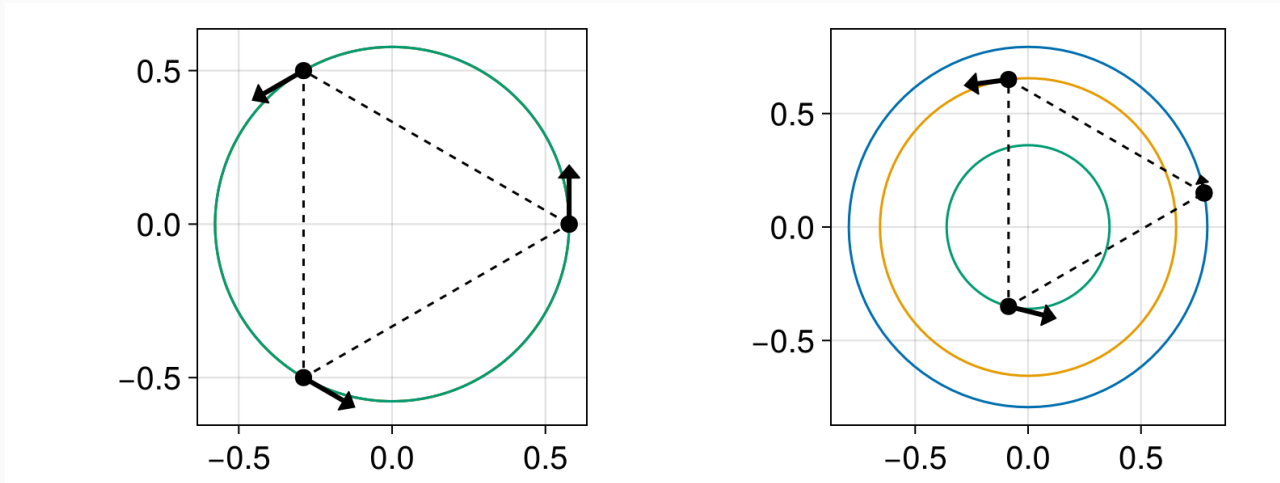
# The third sphere



# The Lagrange relative equilibrium



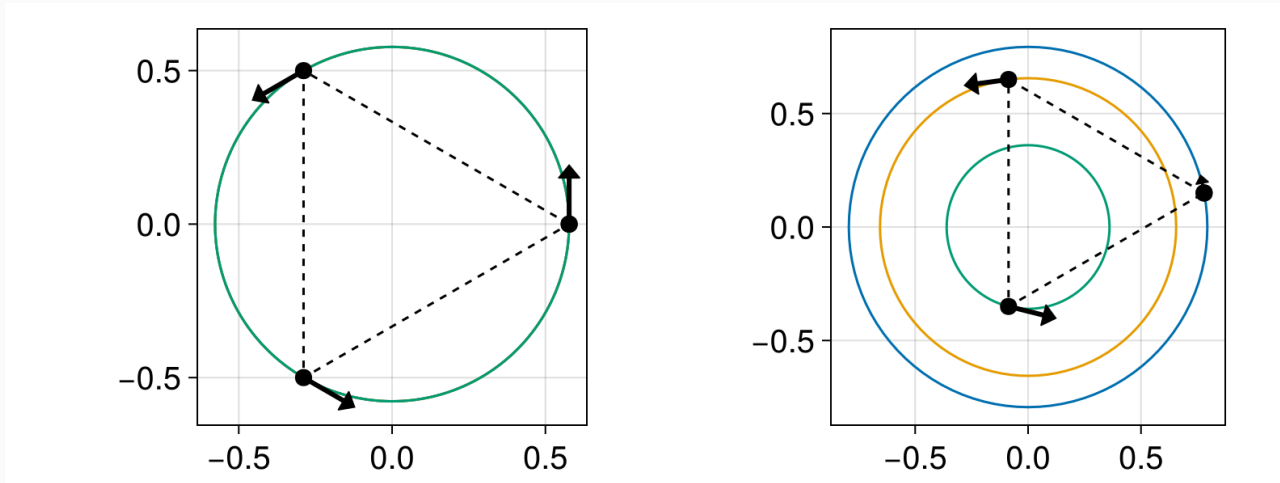
# The Lagrange relative equilibrium



Stable (Gascheau-Routh criterium) if

$$\frac{m_0 m_1 + m_0 m_2 + m_1 m_2}{(m_0 + m_1 + m_2)^2} > \frac{1}{27}$$

# The Lagrange relative equilibrium

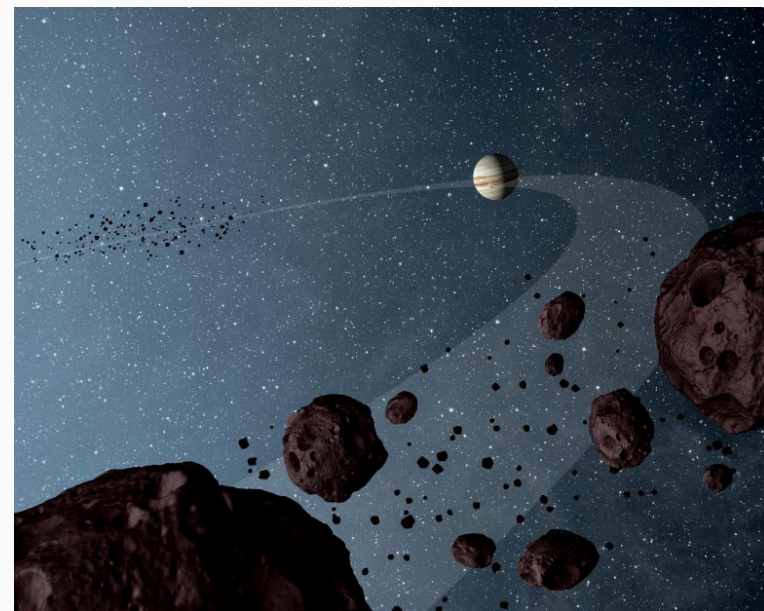
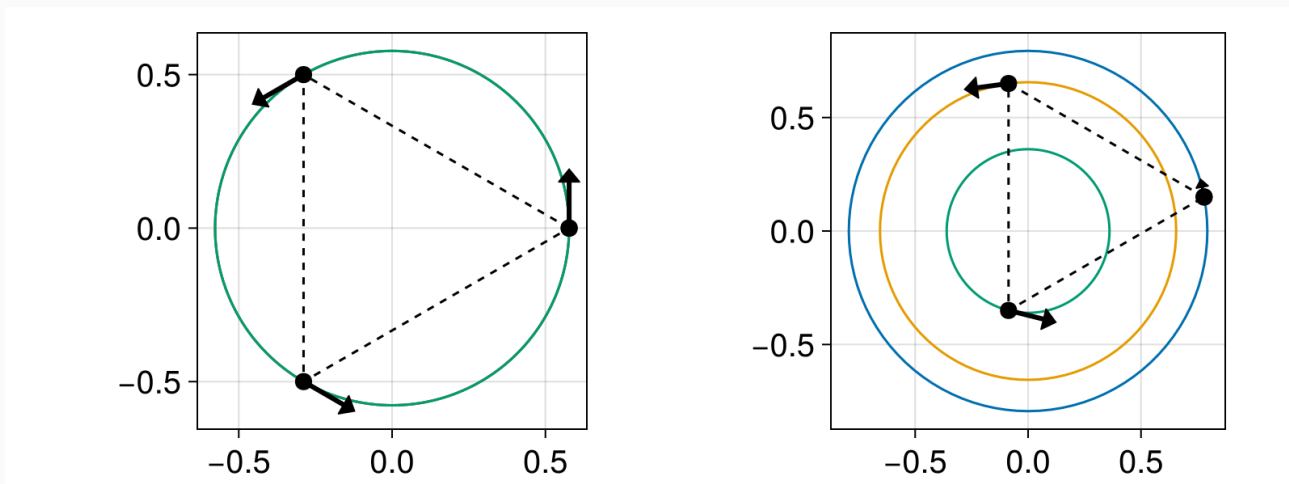


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$$\begin{cases} (1 - \varepsilon, \varepsilon, 0) \rightarrow \varepsilon_G \approx 0.0385 \\ (1 - 2\varepsilon, \varepsilon, \varepsilon) \rightarrow \varepsilon_G \approx 0.0191 \end{cases}$$

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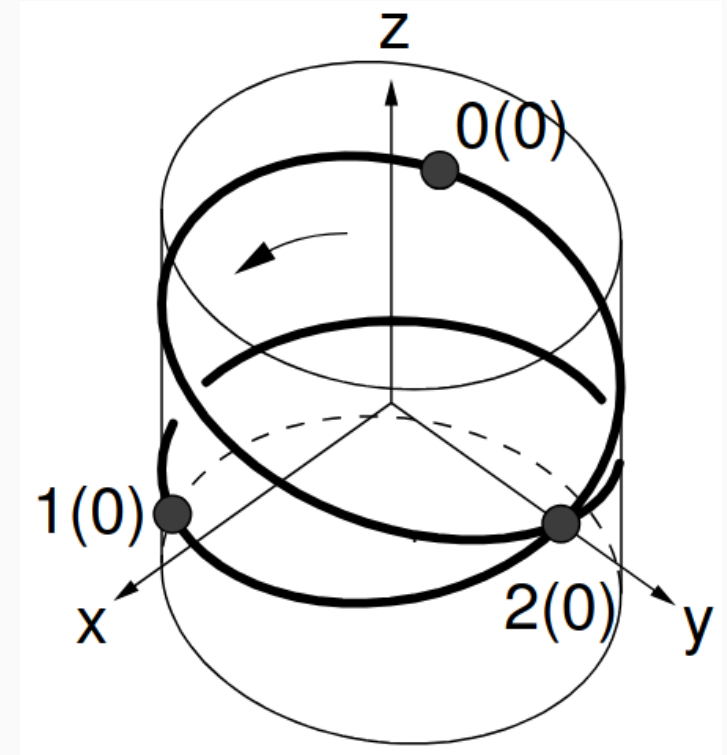
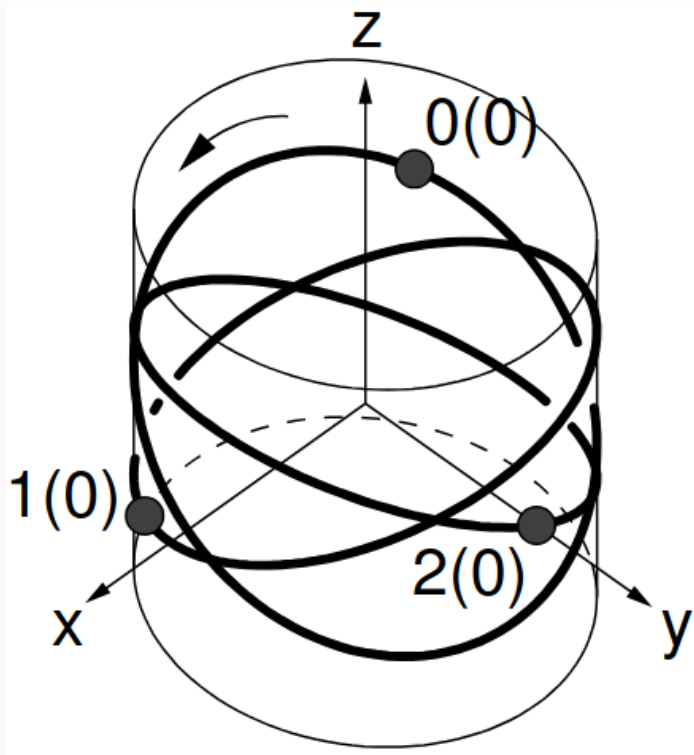
$$\begin{cases} (1 - \varepsilon, \varepsilon, 0) \rightarrow \varepsilon_G \approx 0.0385 \\ (1 - 2\varepsilon, \varepsilon, \varepsilon) \rightarrow \varepsilon_G \approx 0.0191 \end{cases}$$

$\mathcal{VF}_{\mathcal{L}}$ : **state of the art**

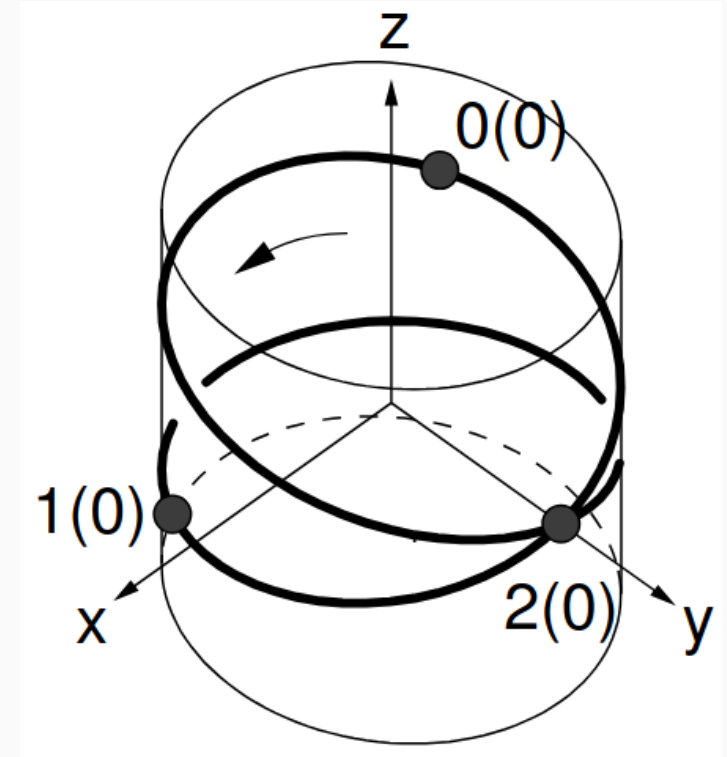
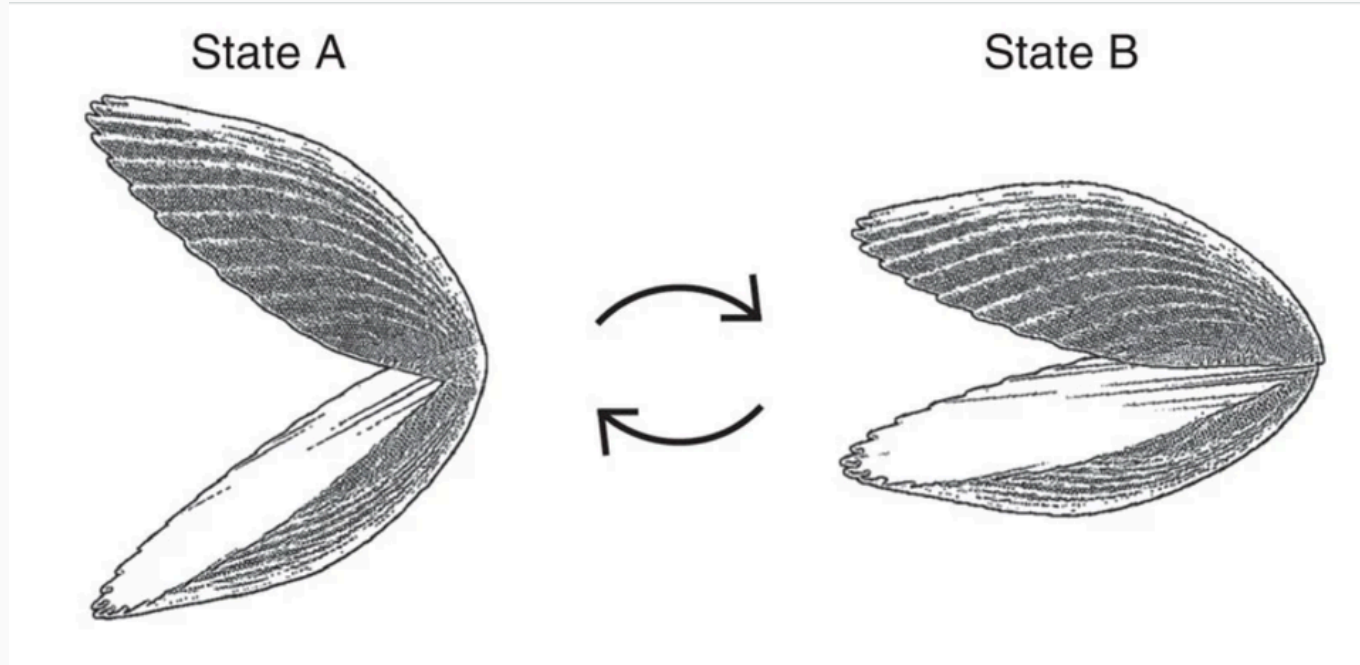
*A peculiar family*

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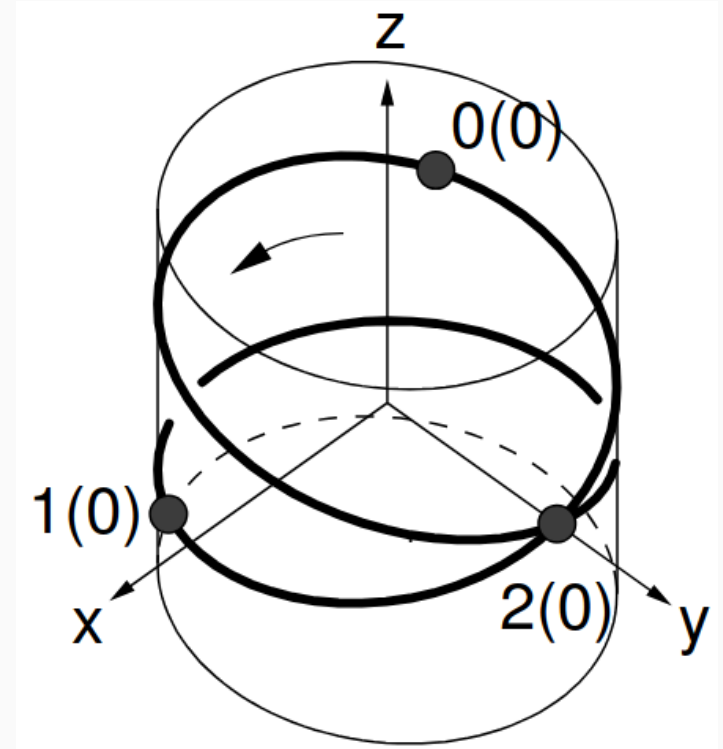
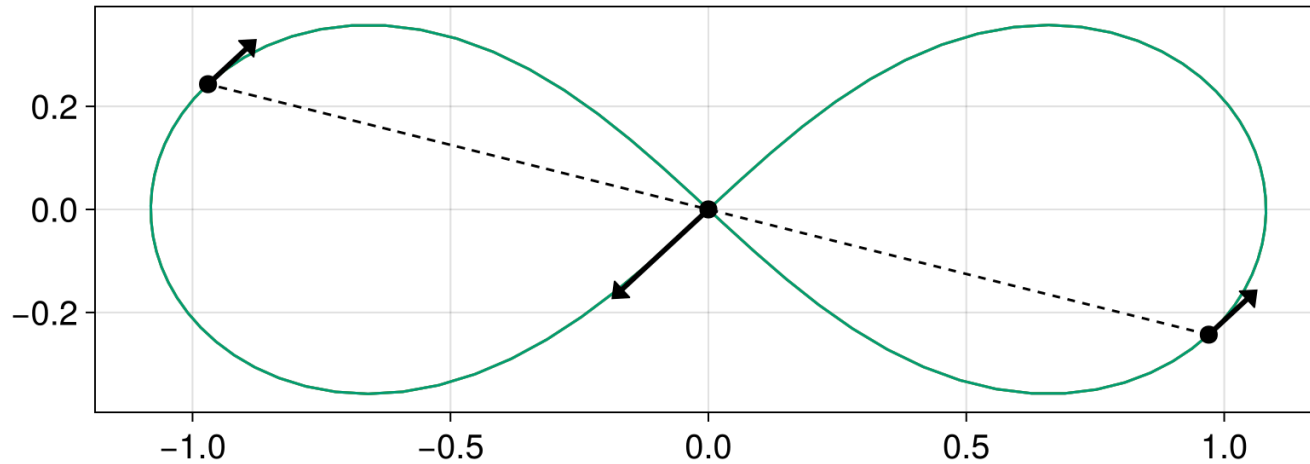
# The vertical family – equal masses



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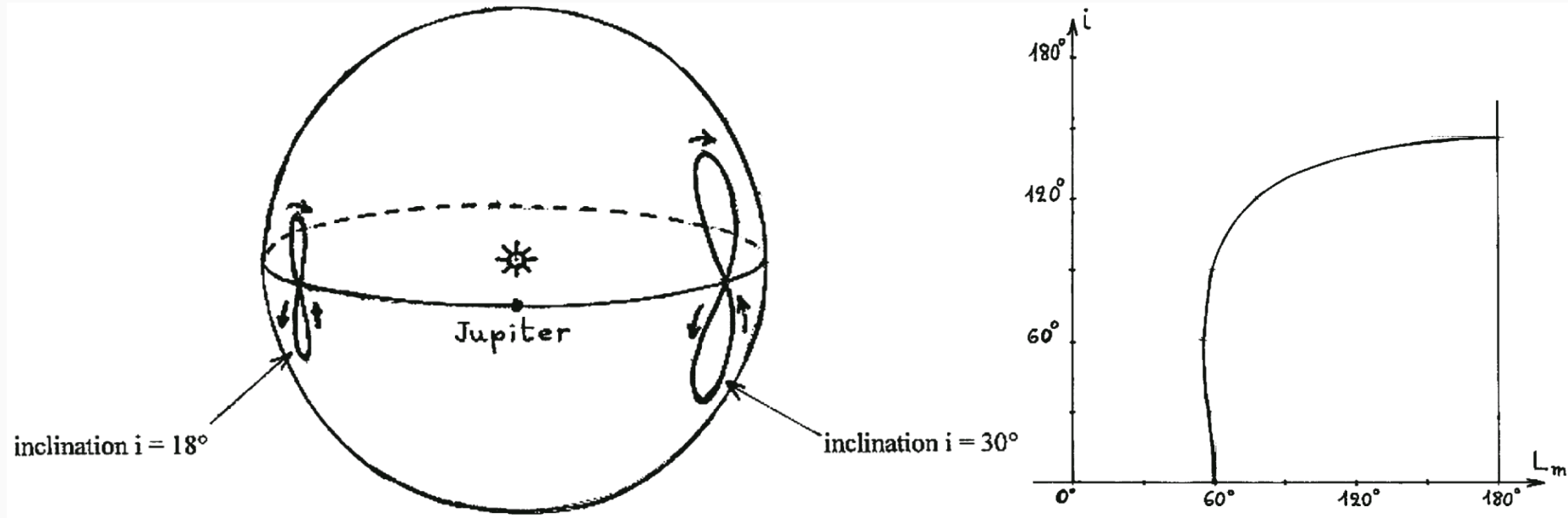


# The vertical family – equal masses



Link conjectured by Marchal in 2000, proved in (Chenciner and Fejoz 2008; Chenciner et al. 2005; Calleja et al. 2024)

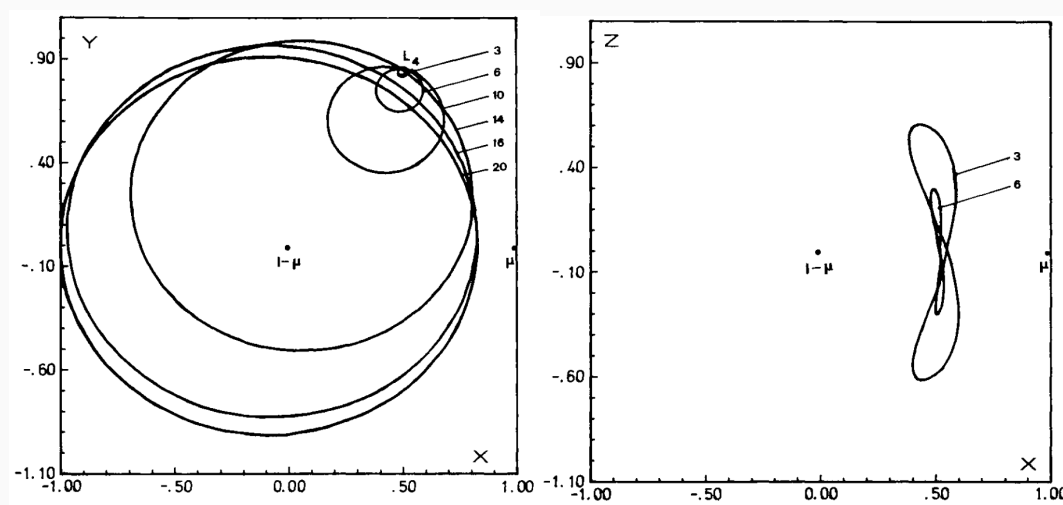
# The vertical family – small masses



- (Marchal 2009) –  $\mathcal{VF}_{\mathcal{L}}$  in the average circular restricted problem

# A historical note: forgetting previous work

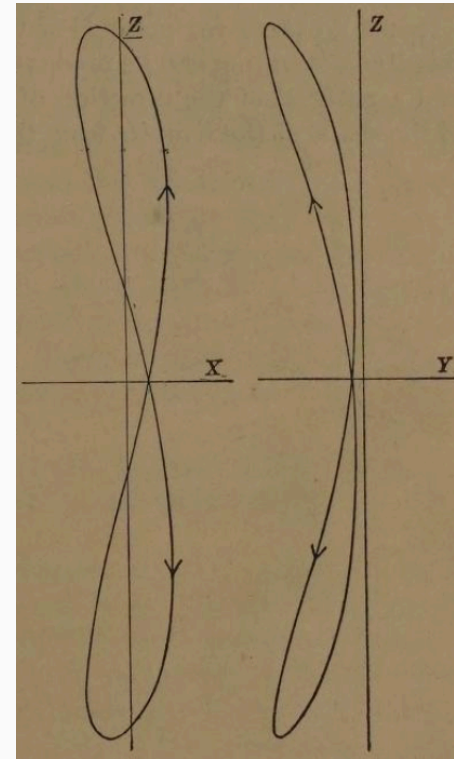
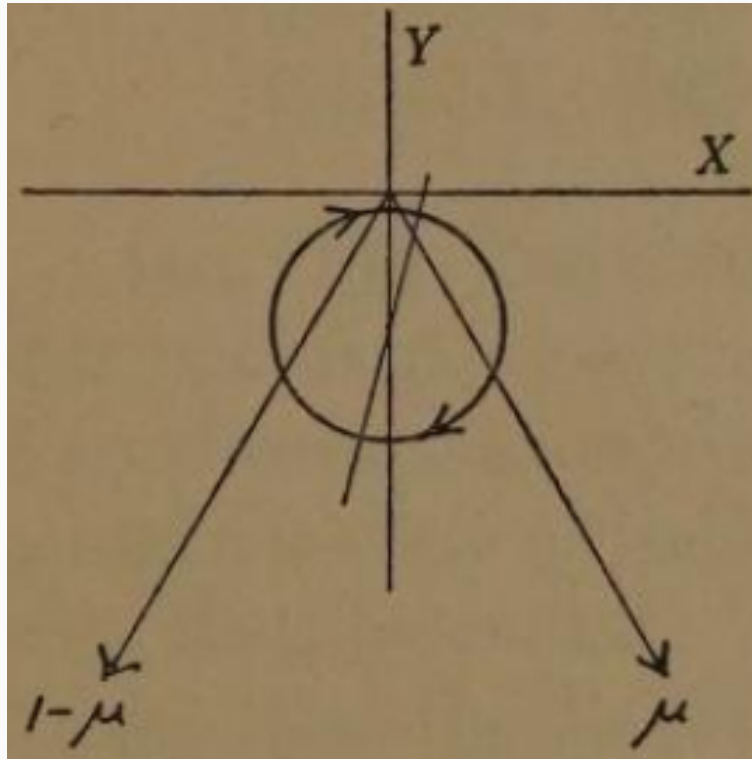
(Marchal 2009) results closely related to (Zagouras 1985) (non-average case)



But (Zagouras 1985)'s method is described in “Zagouras, C.G.: 1982, Post doctoral dissertation, University of Patras, Greece.”

# A historical note: forgetting previous work

... And (Zagouras 1985) is an extension of (Moulton 1920), chap IX (by Buck)!



# Marchal's generalized conjecture

These periodic orbits belong to a three parameter family of similar periodic orbits of the general three body problem, two parameters being the mass ratios of the three bodies and the third a generalization of the inclination  $i$ . [...] The one parameter family  $P_{12}$  of (Marchal 2001) is another limit case of this three parameter family: the case of three equal masses.

— (Marchal 2009)

# An analytical approach by Robutel

Complex Poincaré-like coordinates, heliocentric rotating frame

Planetary problem

$$(1 - \varepsilon(m_1 + m_2), \varepsilon m_1, \varepsilon m_2), \varepsilon \ll 1$$

$\sqrt{\varepsilon}$ -neighborhood of co-orbital motions

$$a_1, a_2 = a^* + \mathcal{O}\left(\varepsilon^{\frac{1}{2}}\right)$$

Averaging over fast angle  $\zeta_2$

The submanifold of eccentricity 0 is invariant under the flow: circular problem

# An analytical approach by Robutel

$$\bar{F}(Z_1, \zeta_1, C) = \alpha Z_1^2 + \varepsilon \beta F_1(\zeta_1, C)$$

Mass  
dependencies

Mass  
independant

For a fixed  $C_0$ , one-parameter family of equilibrium points satisfying

$$Z_1 = 0, \quad \zeta_1 = \zeta(C_0), \quad \frac{\partial F_1}{\partial \zeta_1}(\zeta(C_0), C_0) = 0$$

$F_1$  independant of masses!  $\Rightarrow$  Study valid for all small masses

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- Equilibria in the average rotating problem
- Periodic orbits in the non-average rotating problem
- Quasi-periodic (two frequencies) in the rest frame!

# An analytical approach by Robutel

Stability:  $\mathcal{VF}_{\mathcal{L}}$  is stable in the average circular planetary problem

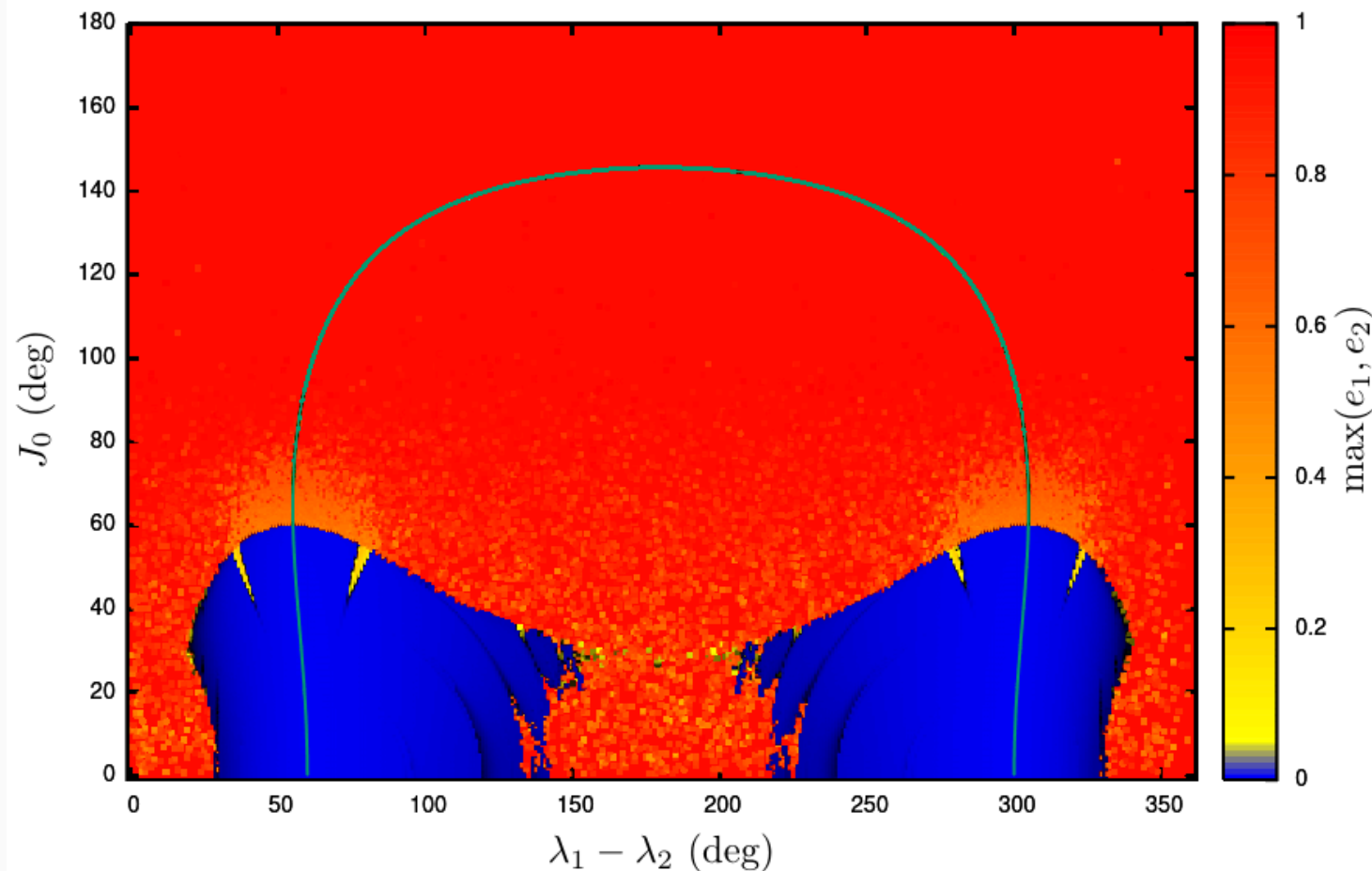


Figure 13: Masses  $(0.998, 10^{-3}, 10^{-3})$

# An analytical approach by Robutel

Stability:  $\mathcal{VF}_{\mathcal{L}}$  is stable in the average **circular** planetary problem

Instability at  $\sim 60^\circ$  seen by (Leleu 2016)

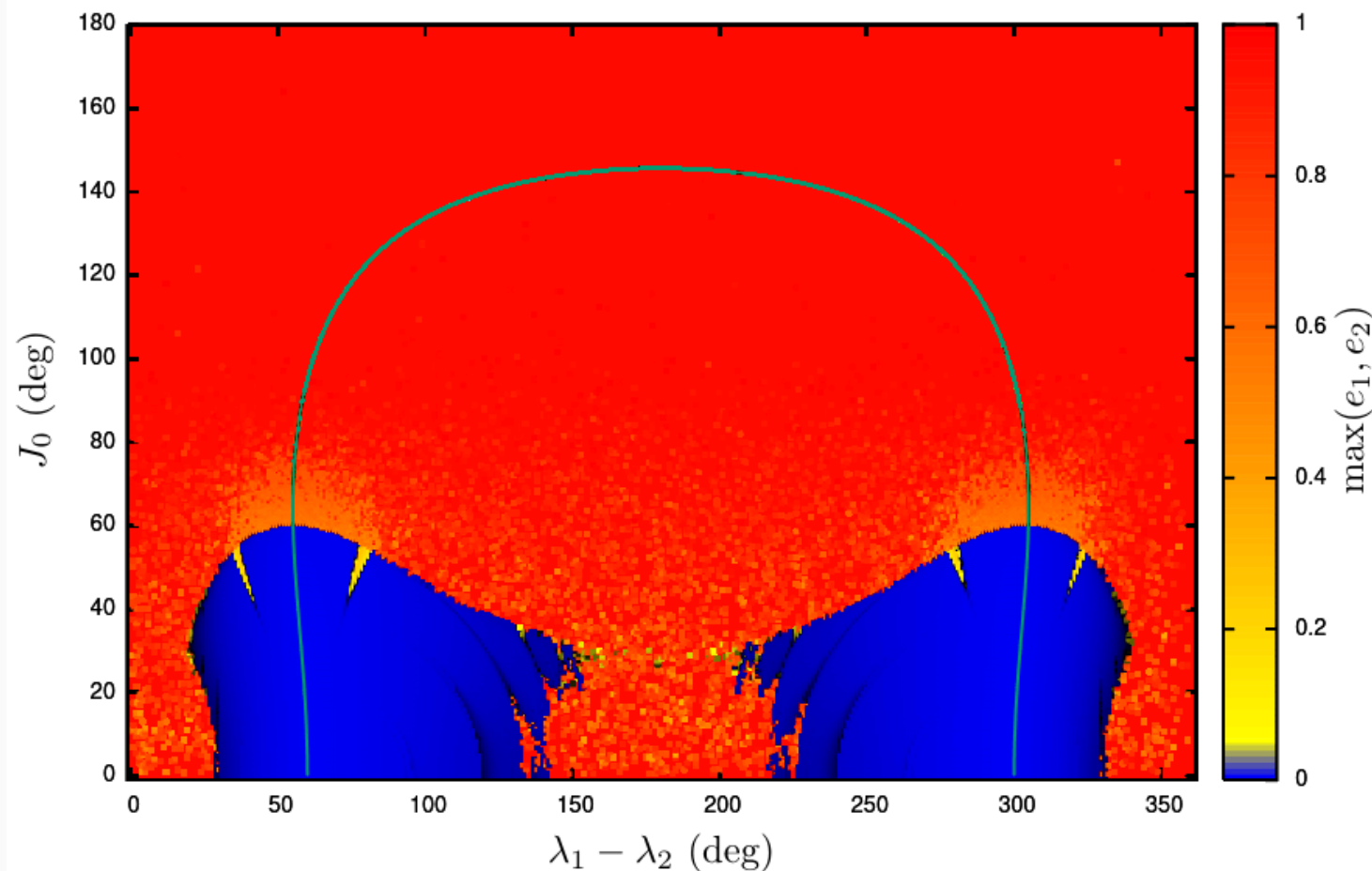


Figure 14: Masses  $(0.998, 10^{-3}, 10^{-3})$

# Numerical methods

*Searching for recurring  
dynamics*



# Manipulating the three-body problem

$$\begin{cases} \mathbf{x} \\ f : \mathbf{x} \mapsto \dot{\mathbf{x}} \end{cases}$$

# Manipulating the three-body problem

$$\begin{cases} \mathbf{x} = \left( \overbrace{\mathbf{q}_0, \mathbf{q}_1, \mathbf{q}_2}^{\text{positions}}, \underbrace{\mathbf{p}_0, \mathbf{p}_1, \mathbf{p}_2}_{\text{momenta}} \right) \\ f : \mathbf{x} \mapsto \dot{\mathbf{x}} = (\dot{\mathbf{q}}_0, \dot{\mathbf{q}}_1, \dot{\mathbf{q}}_2, \dot{\mathbf{p}}_0, \dot{\mathbf{p}}_1, \dot{\mathbf{p}}_2) \end{cases}$$

$$\dot{\mathbf{q}}_i = \frac{\mathbf{p}_i}{m_i}, \dot{\mathbf{p}}_i = \sum_{j \neq i} \mathcal{G} m_i m_j \frac{\mathbf{q}_j - \mathbf{q}_i}{\|\mathbf{q}_j - \mathbf{q}_i\|^3}$$

# Periodic orbits

The flow

$$\Phi_t : \mathbf{x}_0 \mapsto \mathbf{x}(t) = \int_0^t f \cdot \mathbf{x}$$

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A **periodic orbit**

$$\Phi_T(\mathbf{x}) = \mathbf{x}$$

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We search for zeros of  $\Phi_T - \text{id}$

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A **periodic orbit**

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We search for zeros of  $\Phi_T - \text{Id}$

Two problems

- Compute  $\Phi_T$
- Find the zeros

# Other approaches to search of periodic orbits

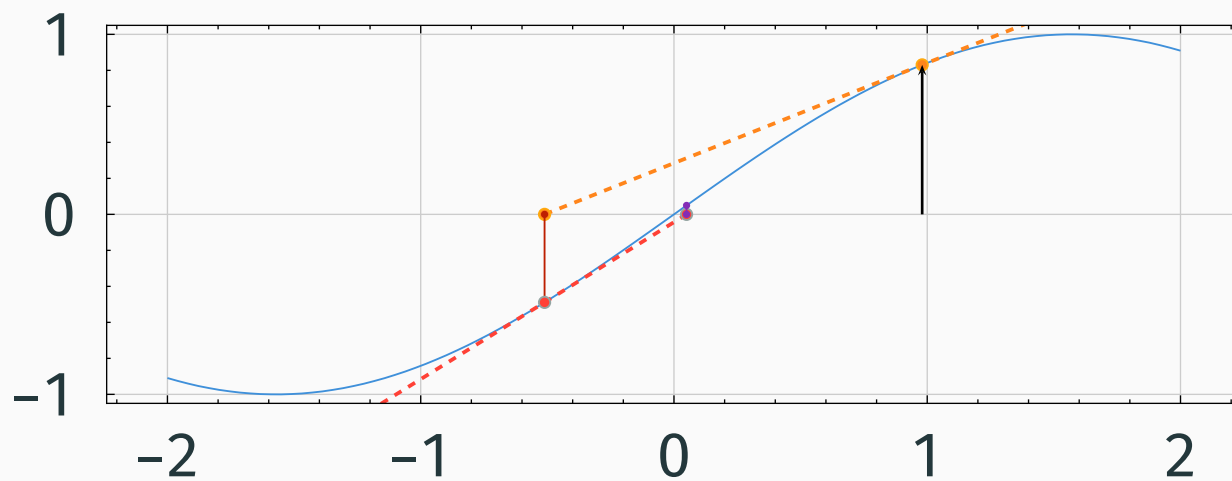
- Searching for action extrema
  - (Moore 1993)...
- Iterations on quasi-periodic approximations with frequency analysis
  - (Laskar 1990; Couetdic et al. 2010)...
- Lindstedt-Poincaré methods
  - (Zagouras 1985)...

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- Lindstedt-Poincaré methods
  - (Zagouras 1985)...
- Newton method for fixed point of the flow
  - (Hénon 1974; 1997)

# Looking for zeros

Multi-dimensional  
Newton-Raphson method

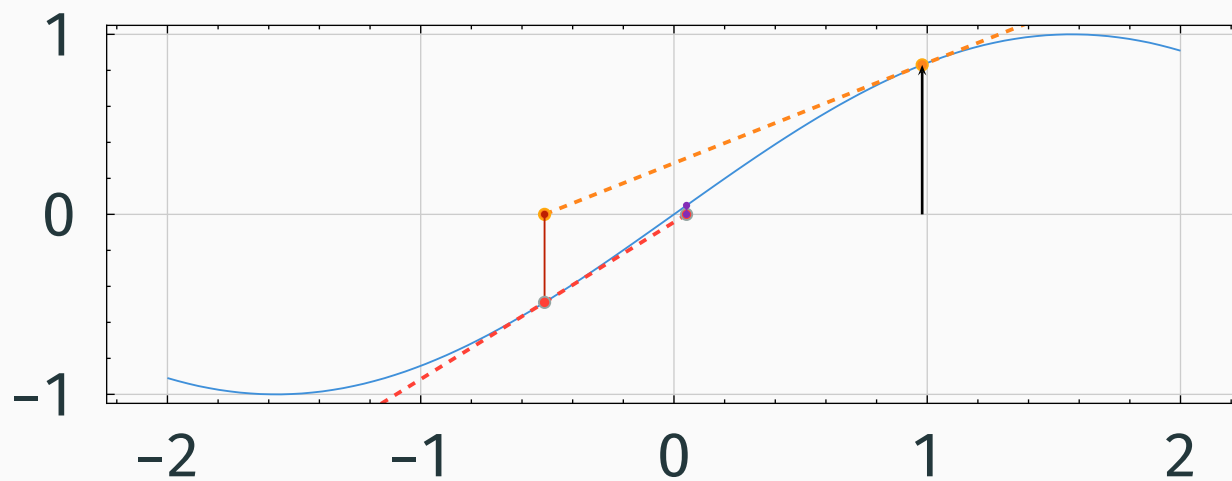


# Looking for zeros

Multi-dimensional  
**Newton-Raphson** method

Two necessary conditions

- Good initial guess
- 

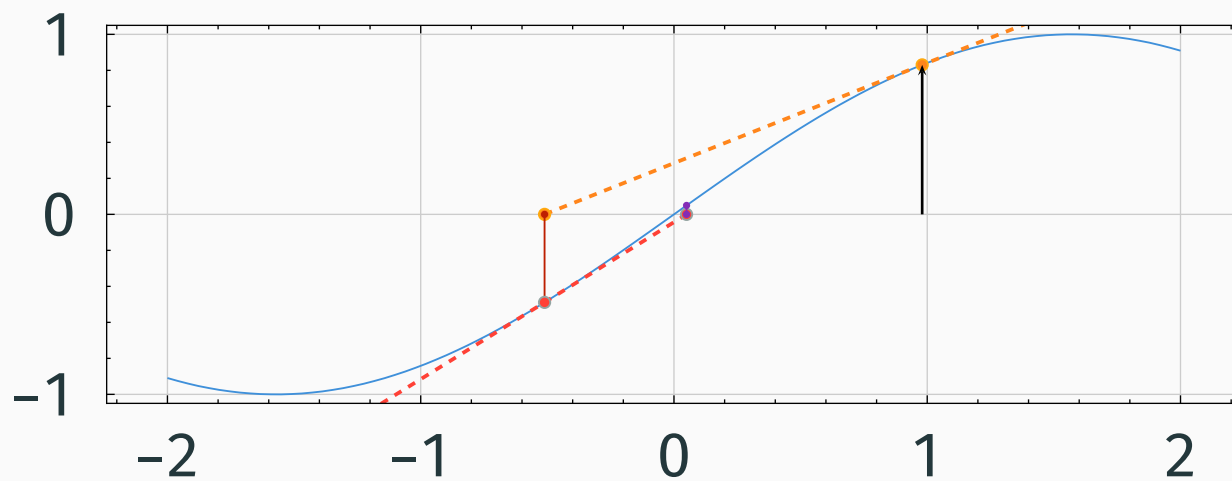


# Looking for zeros

Multi-dimensional  
**Newton-Raphson** method

Two necessary conditions

- Good initial guess
- Local uniqueness



# Retrieving local uniqueness

We add **sections**: we look for zeros of

$$f : \begin{cases} \mathcal{M} \rightarrow \mathcal{M} \times \mathbb{R}^m \\ \mathbf{x} \mapsto \begin{pmatrix} \Phi_T(\mathbf{x}) - \mathbf{x} \\ \sigma(\mathbf{x}) \end{pmatrix} \end{cases}$$

# Retrieving local uniqueness

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$\sigma$  can help us:

- Isolate a point on the orbit
- Isolate an orbit in a family
- etc.

# Numerical integration

- Numerical integration with `taylor` (Jorba and Zou 2005; Gimeno et al. 2022)
  - Why Taylor ? Symplectic integrators: difficult to add variational equations
  - Step size **and** order as parameters: more refined strategies
- Gives us access to the derivatives: necessary for eg. rootfinding, stability
- Integrated in **julia** with `TaylorInterface.jl`

# Why Julia?

 v0.1 in 2012, v1.0 in 2017 (Bezanson et al. 2017)

- Pros:
  - Easy to write in and optimize
  - Significant community in scientific computing
  - Reproducibility
- Cons
  - Smaller community
  - Instability (active development)

```
function dissipative_std_map(input)
    ε = 0.98; λ = 0.9; μ = 0.1
    x, y = input
    y_prime = y + ε * v(x) + λ * (x - μ)
    return (x + y_prime, y_prime)
end

grid = [
    (x, y) for x in LinRange(0., 1., 51),
              y in LinRange(-2., 2., 21)
]
first_iteration = map(
    dissipative_std_map, grid
)
```

# Putting everything together

~ 8400 lines of code of 

- Numerical integration with `TaylorInterface.jl`
- Sections added in Julia, derived with automatic differentiation
- Custom Newton-Raphson implementation

# Planetary results

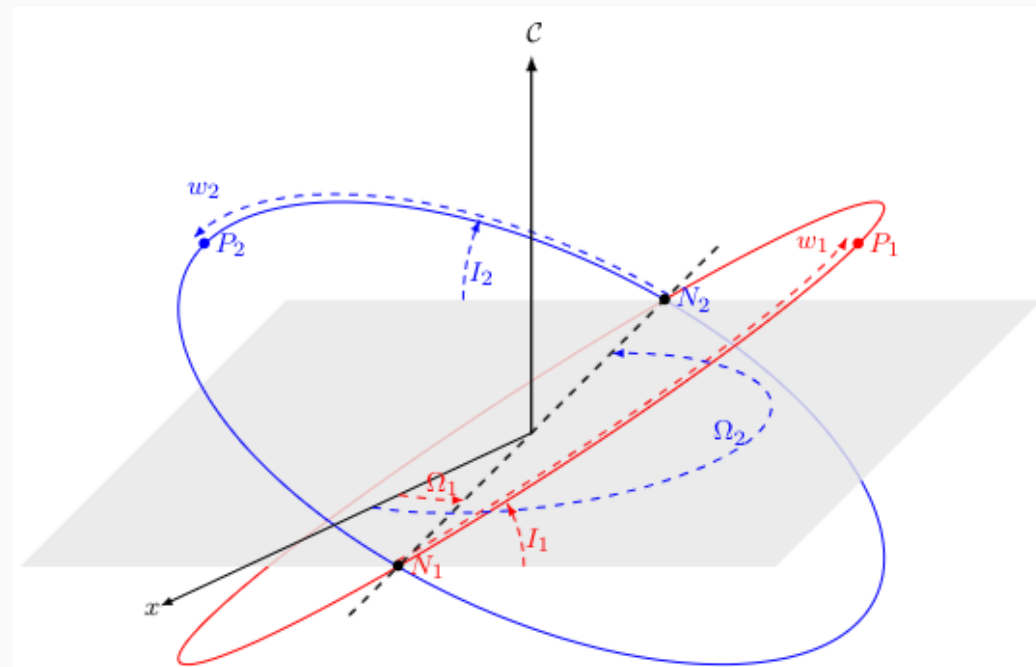
$\mathcal{VF}_{\mathcal{L}}$  at low masses

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# Hill coordinates

## Hill coordinates

- $r_i$  radii,  $\tilde{r}_i$  radial velocities
- $w_i$  draconic true anomalies,  
 $G_i$  angular momenta
- $\Omega_i$  node longitudes,  
 $G_i \cos(i_i)$  vertical angular momenta
- $C$  total angular momentum



# Hill coordinates

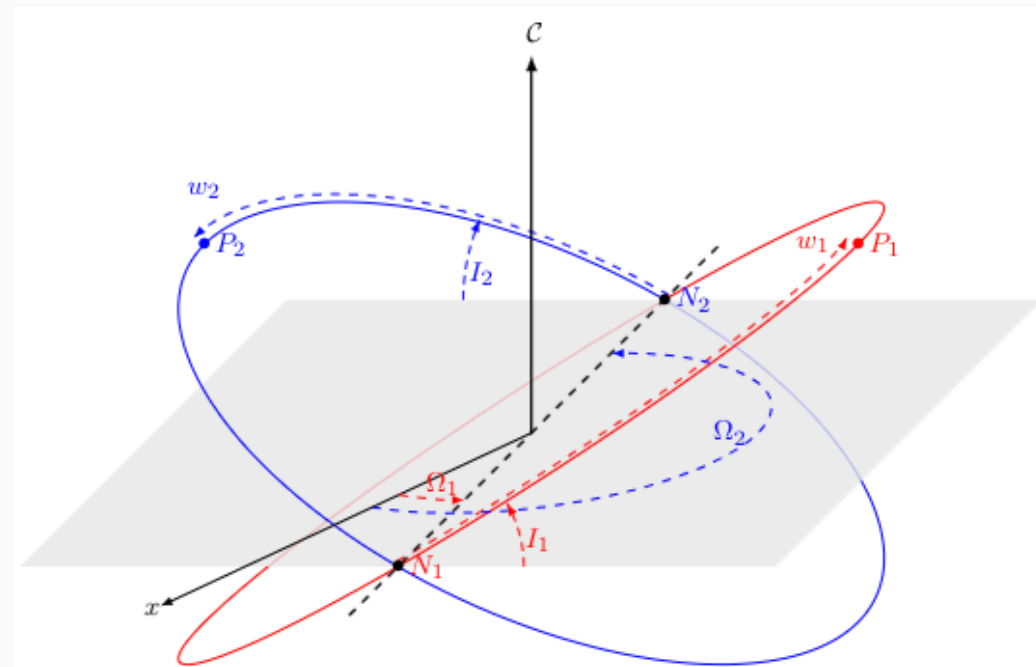
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## Draconic frame

We can recover  $\Omega$ :

$$\dot{\Omega}(t) = \frac{\partial \mathcal{H}}{\partial C}(\mathbf{x}(t))$$



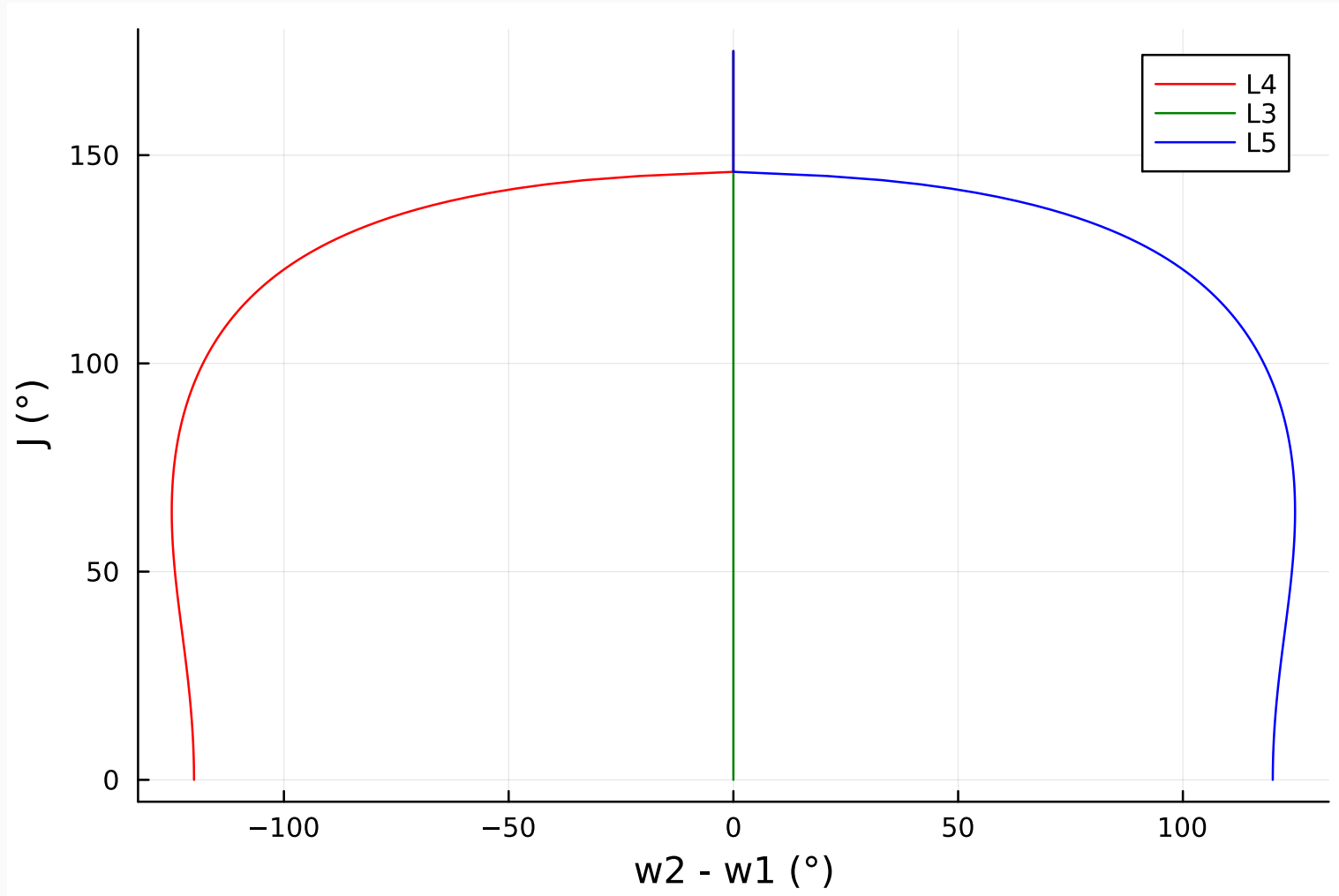
# Numerical setup

We use the numerical  setup described before

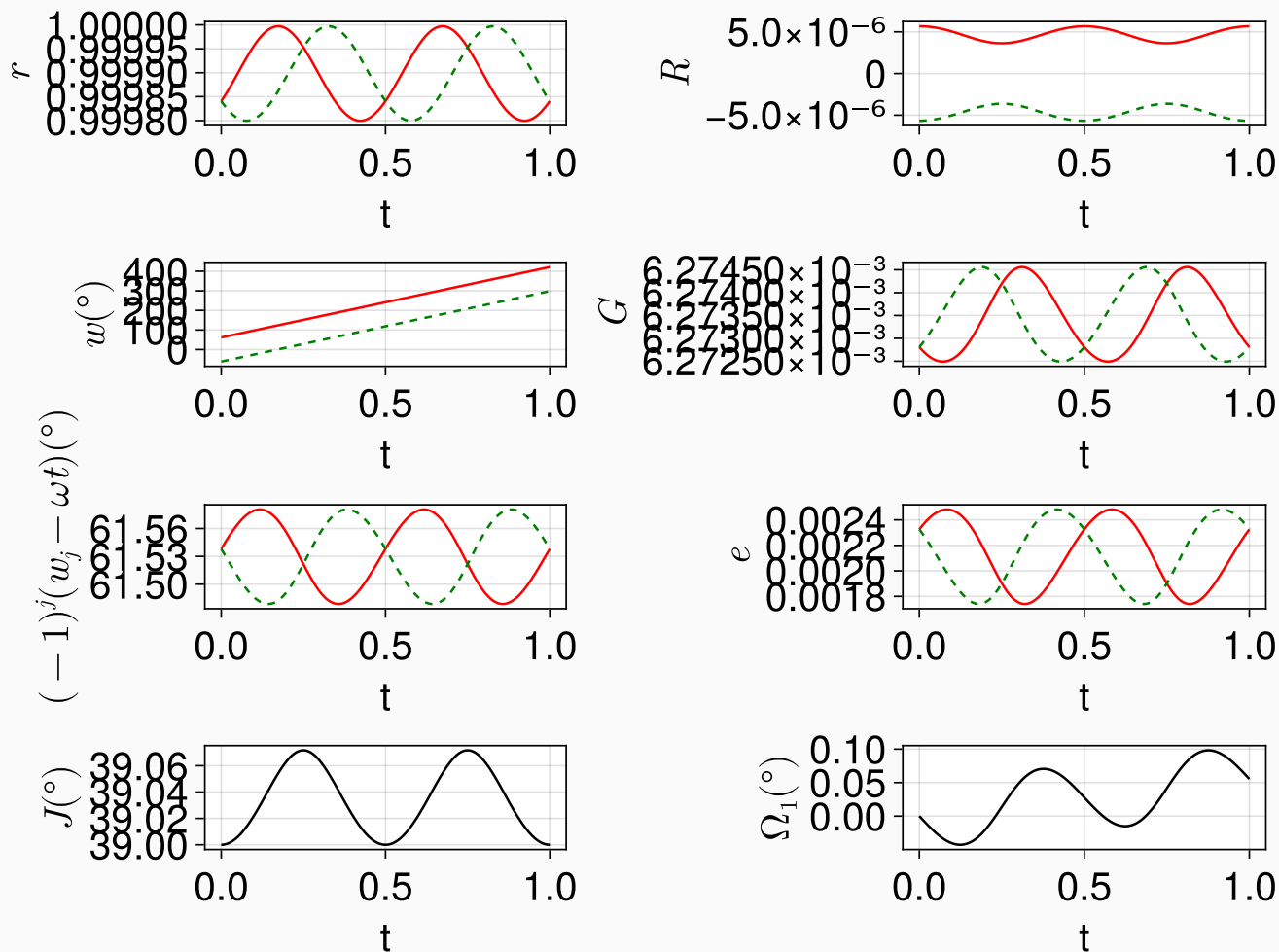
- Reduced Hill coordinates: 8+1 dimensions
- Two equal masses case:  $(1 - 2\varepsilon, \varepsilon, \varepsilon), 0 < \varepsilon \leq \frac{1}{3}$
- Initial guess: Lagrange equilibrium, then previous orbit
- Two sections:

$$\sigma(\mathbf{x}) = \begin{pmatrix} W_1 + W_2 \\ J(\mathbf{x}) - J_0 \end{pmatrix}$$

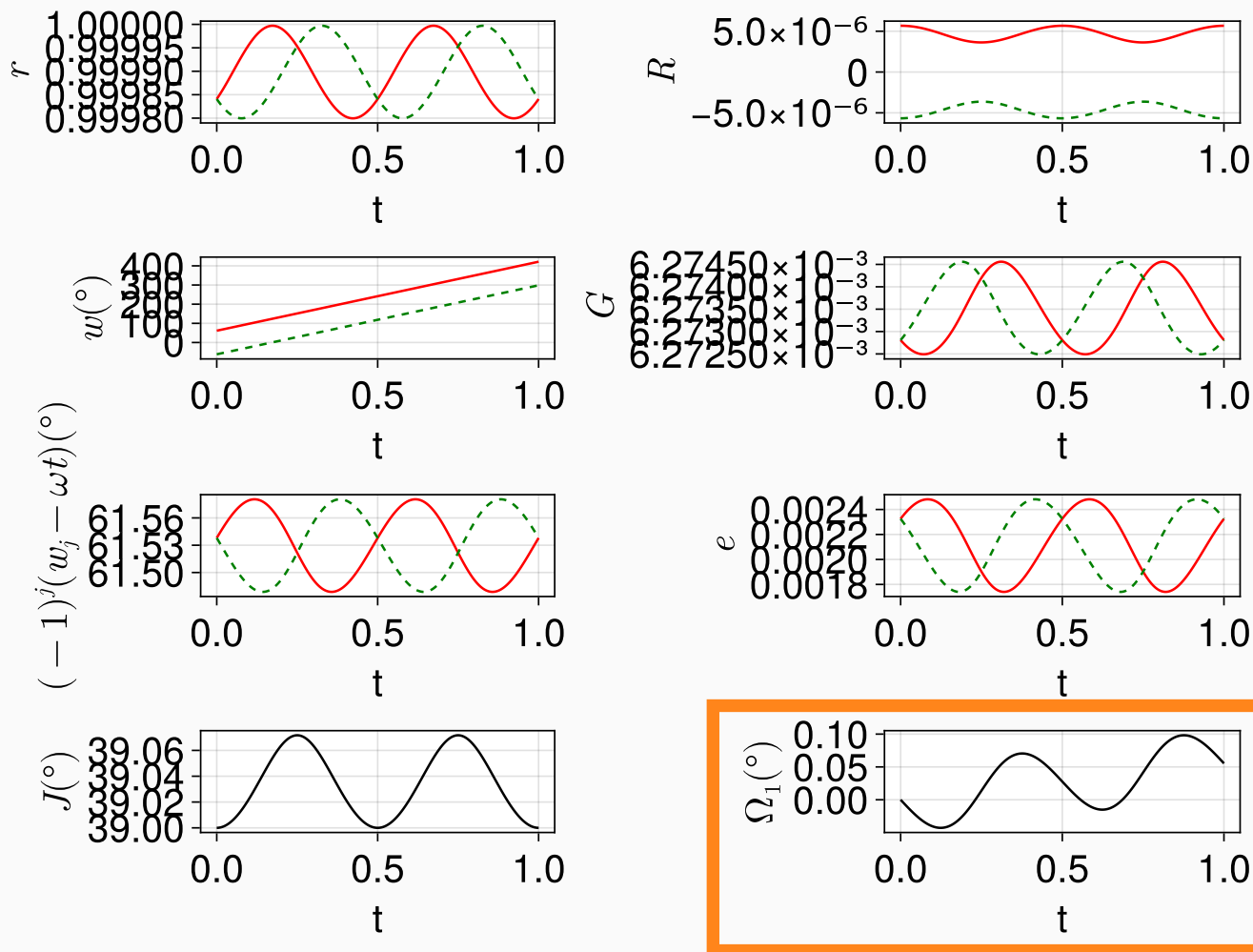
# Small masses: $\varepsilon = 10^{-3}$



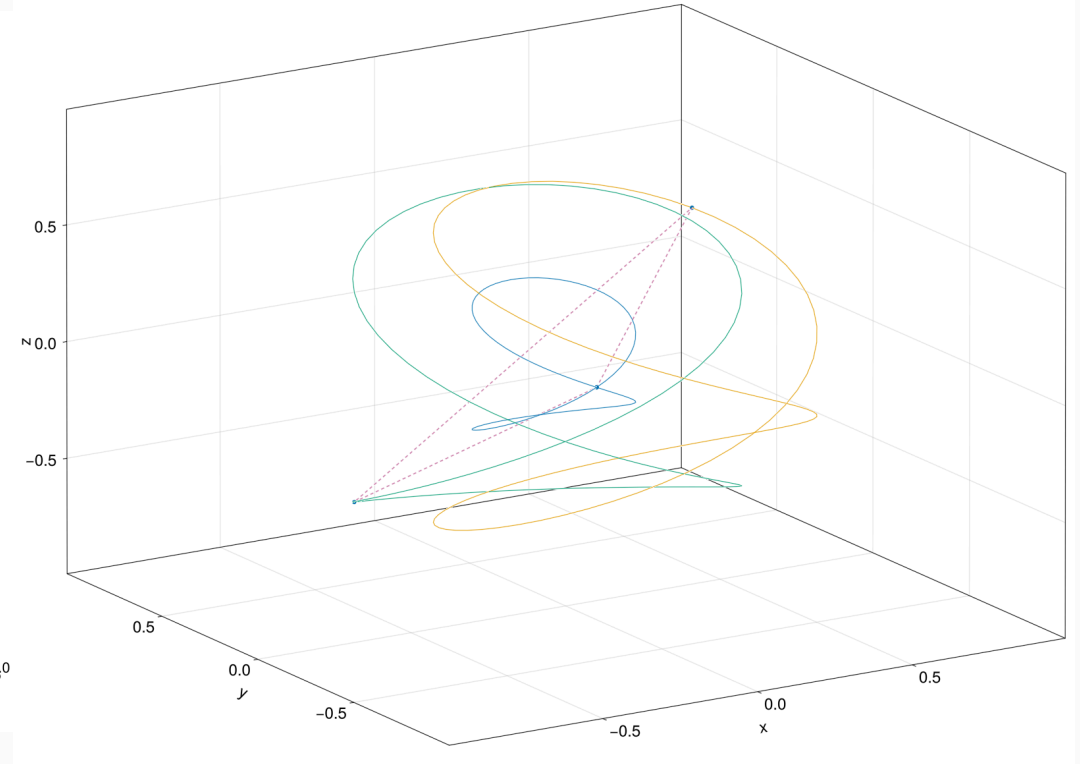
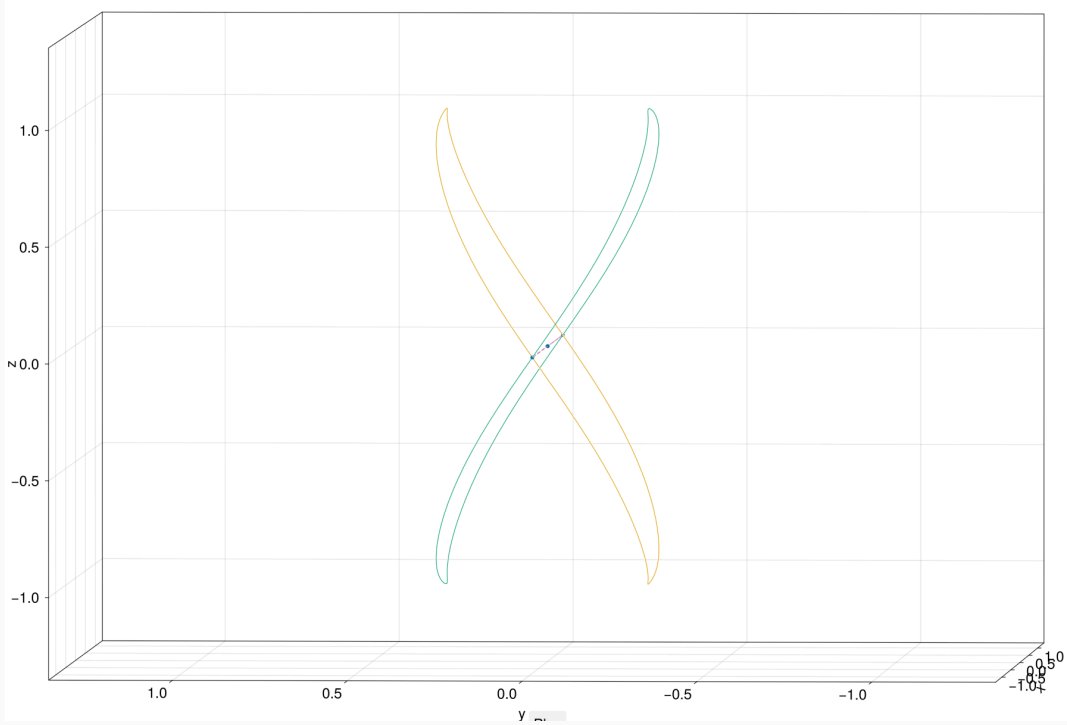
# Coordinates along trajectory at $\varepsilon = 10^{-3}, J = 39^\circ$



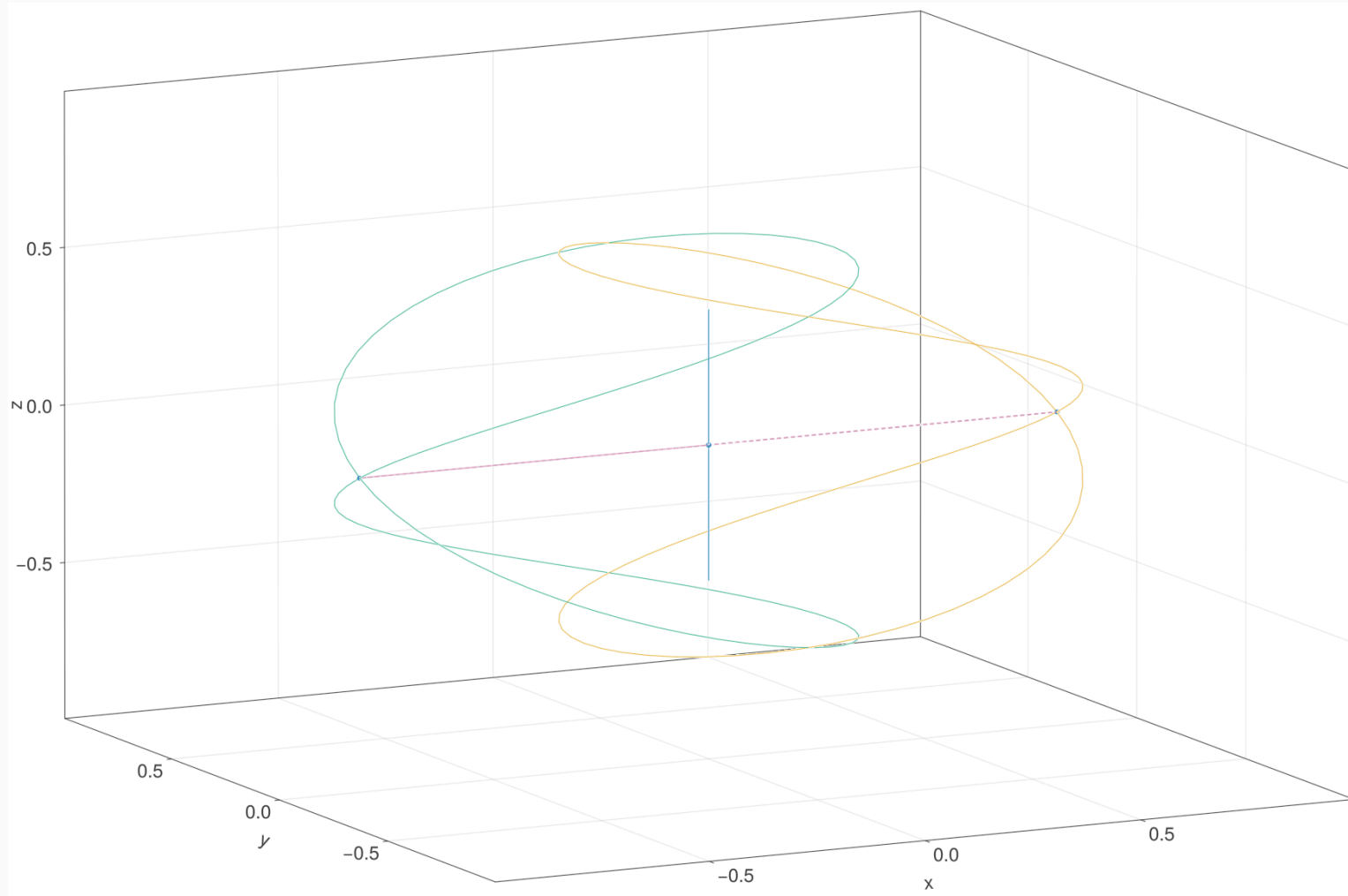
# Coordinates along trajectory at $\varepsilon = 10^{-3}, J = 39^\circ$



# Trajectory examples – $\mathcal{VF}_{\mathcal{L}}$ , $\varepsilon = 0.21, J = 106^\circ$



# Trajectory example – $\mathcal{VF}_\varepsilon$ , $\varepsilon = 0.21, J = 130^\circ$

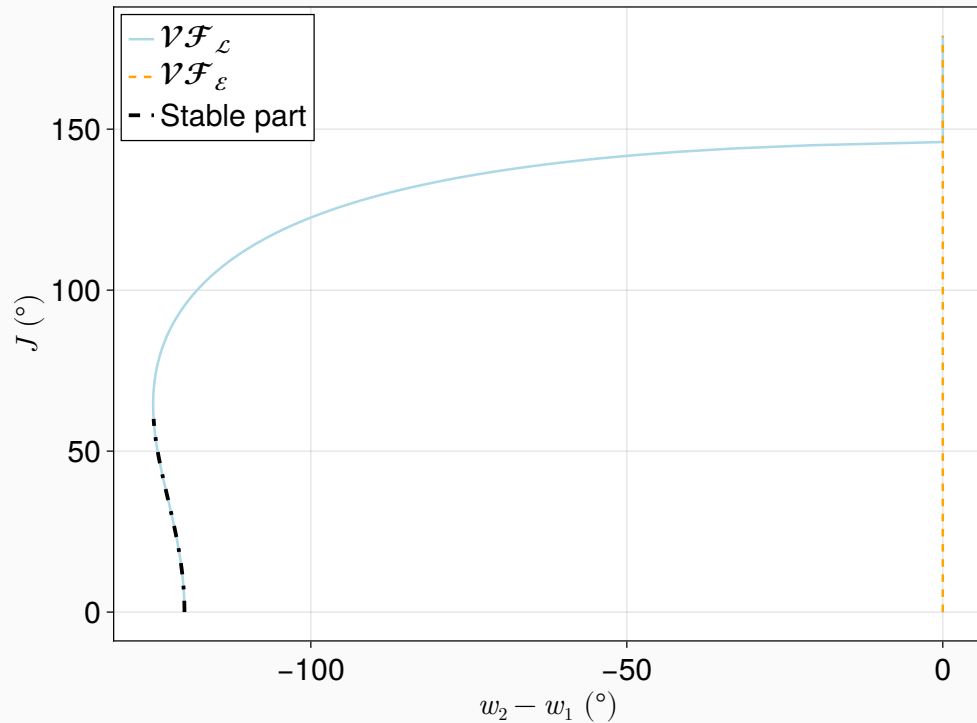


# Stability

*Should I stay or should I go?*

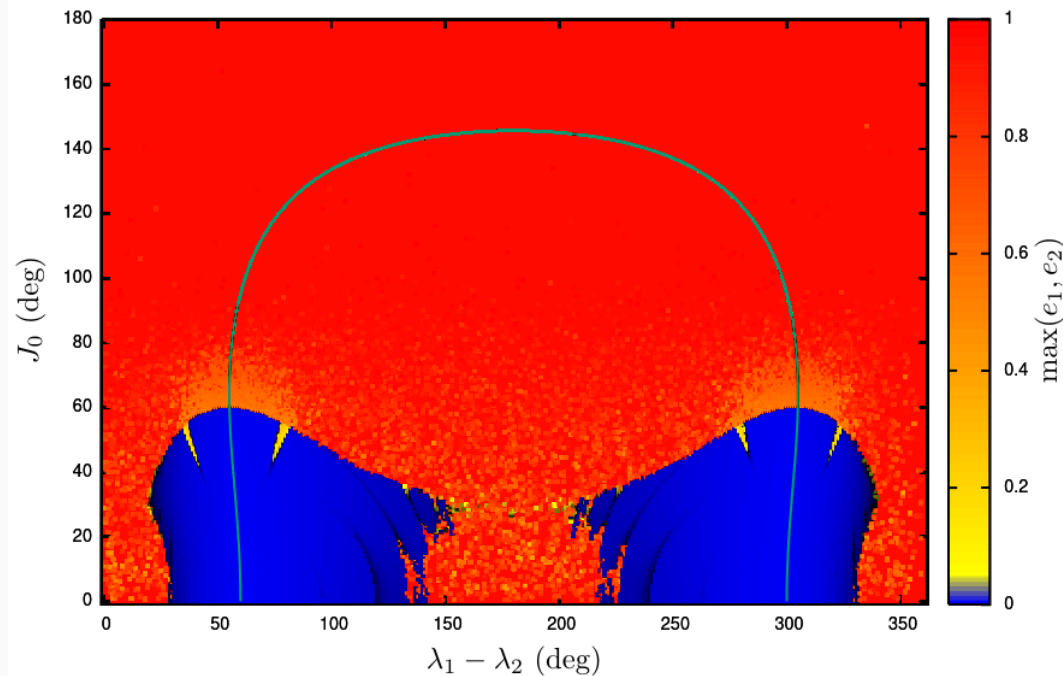
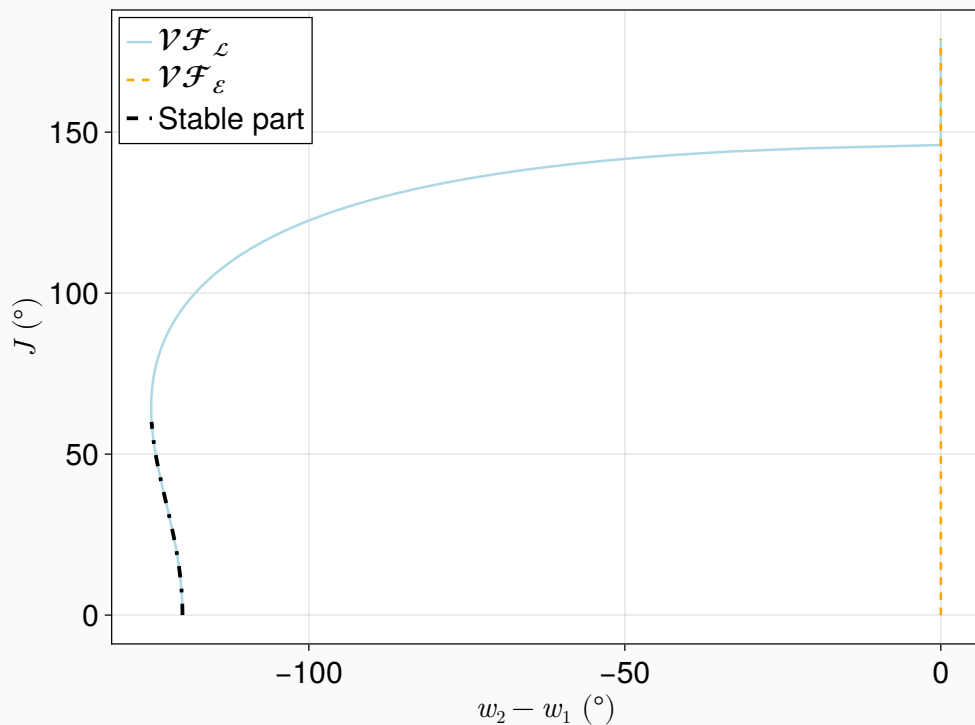


# Linear stability at $\varepsilon = 10^{-3}$



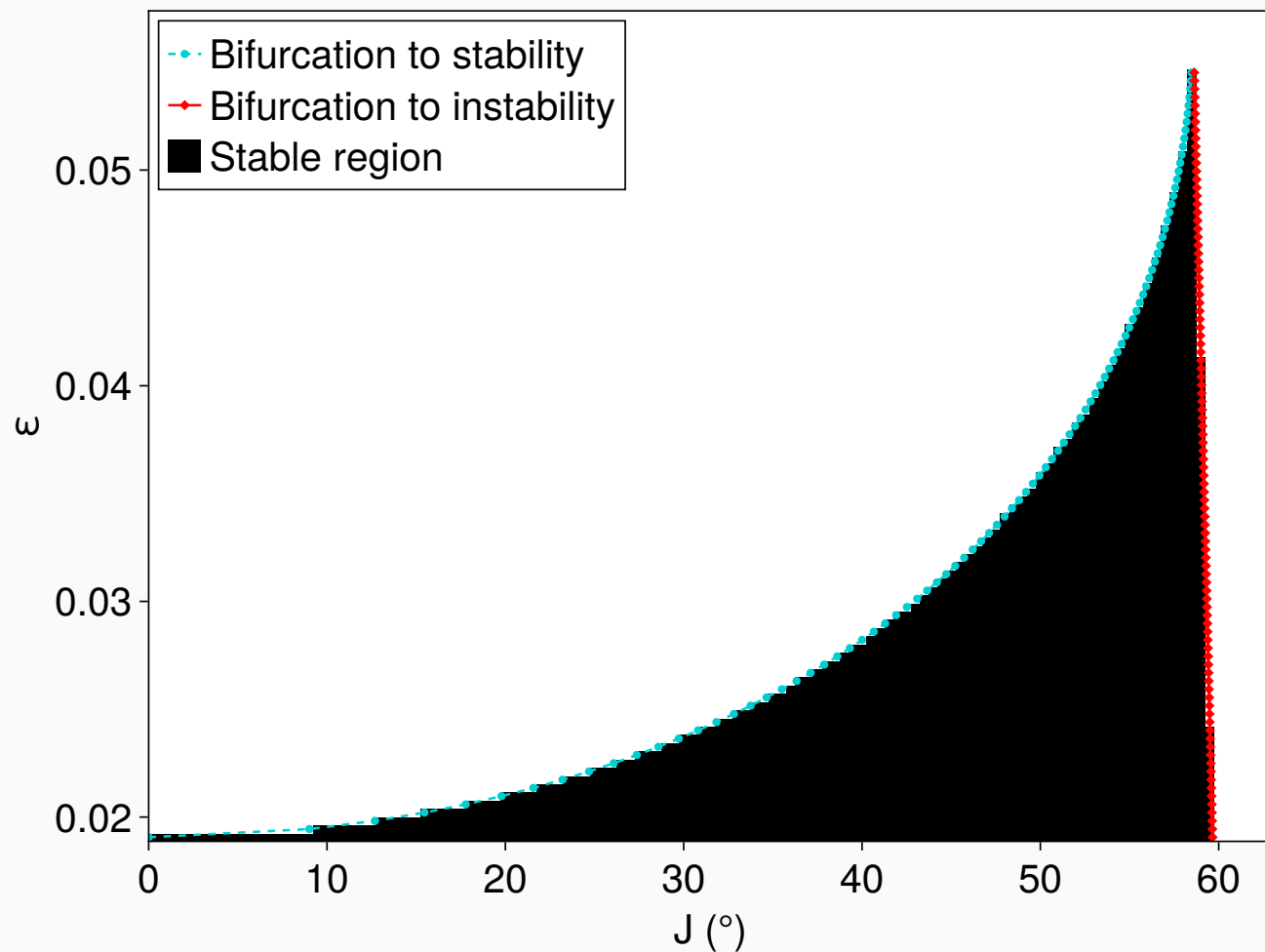
Stable if  $Sp(D\Phi_T) \subset \text{unit circle}$

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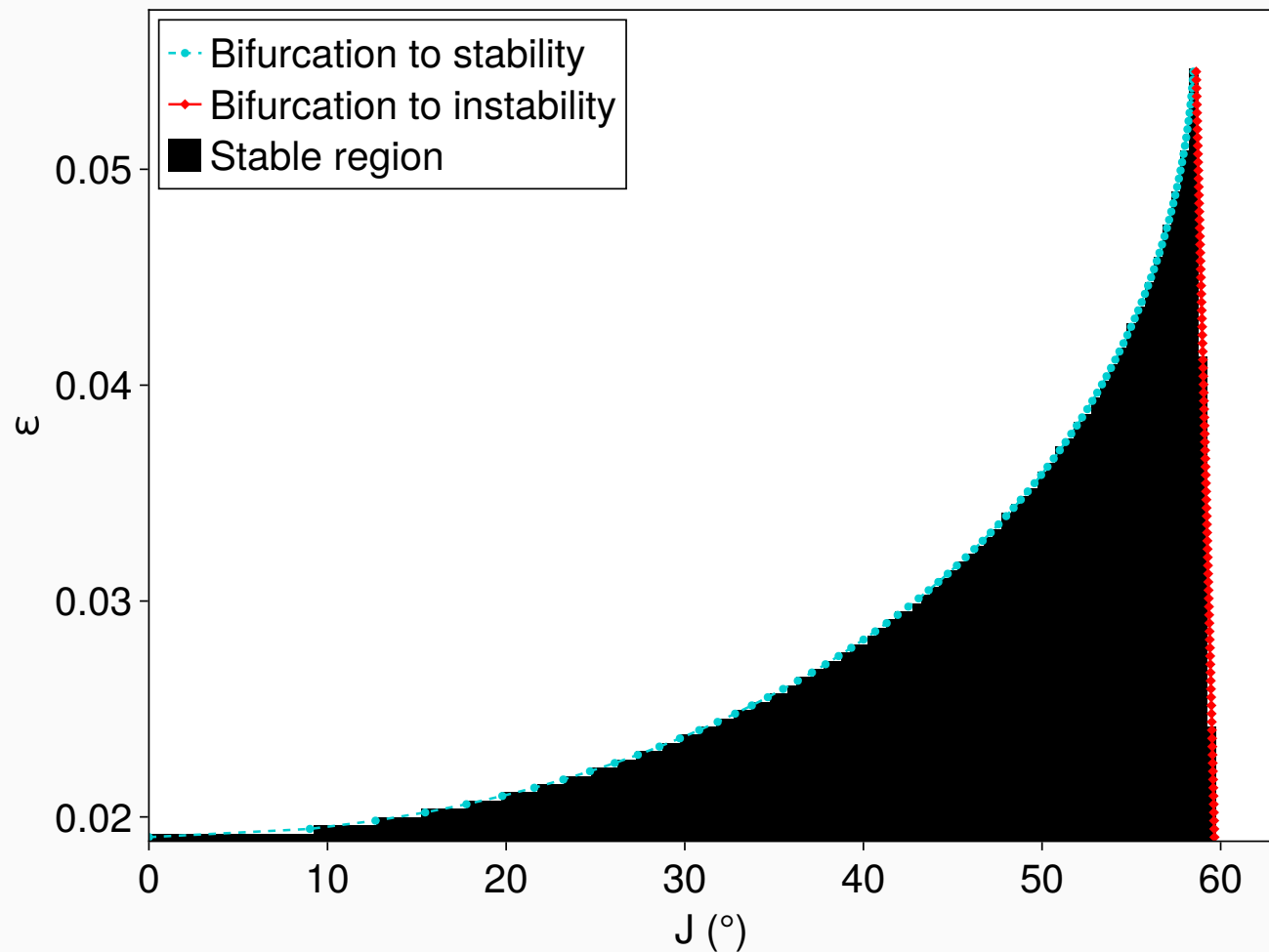


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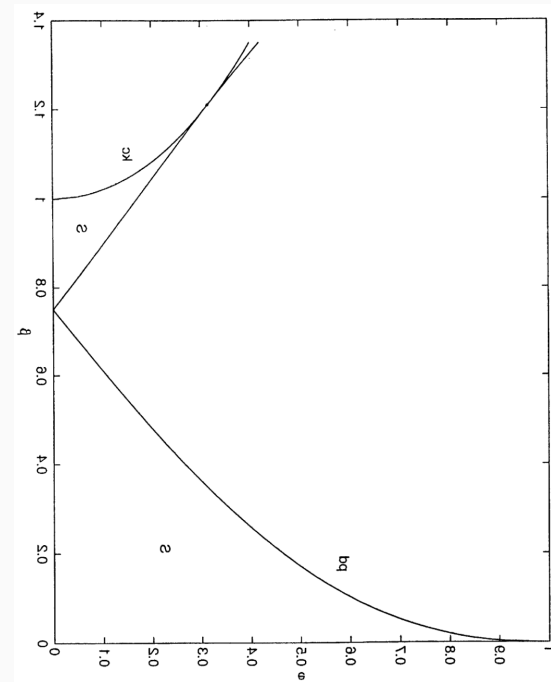
# Increasing masses: above Gascheau's value



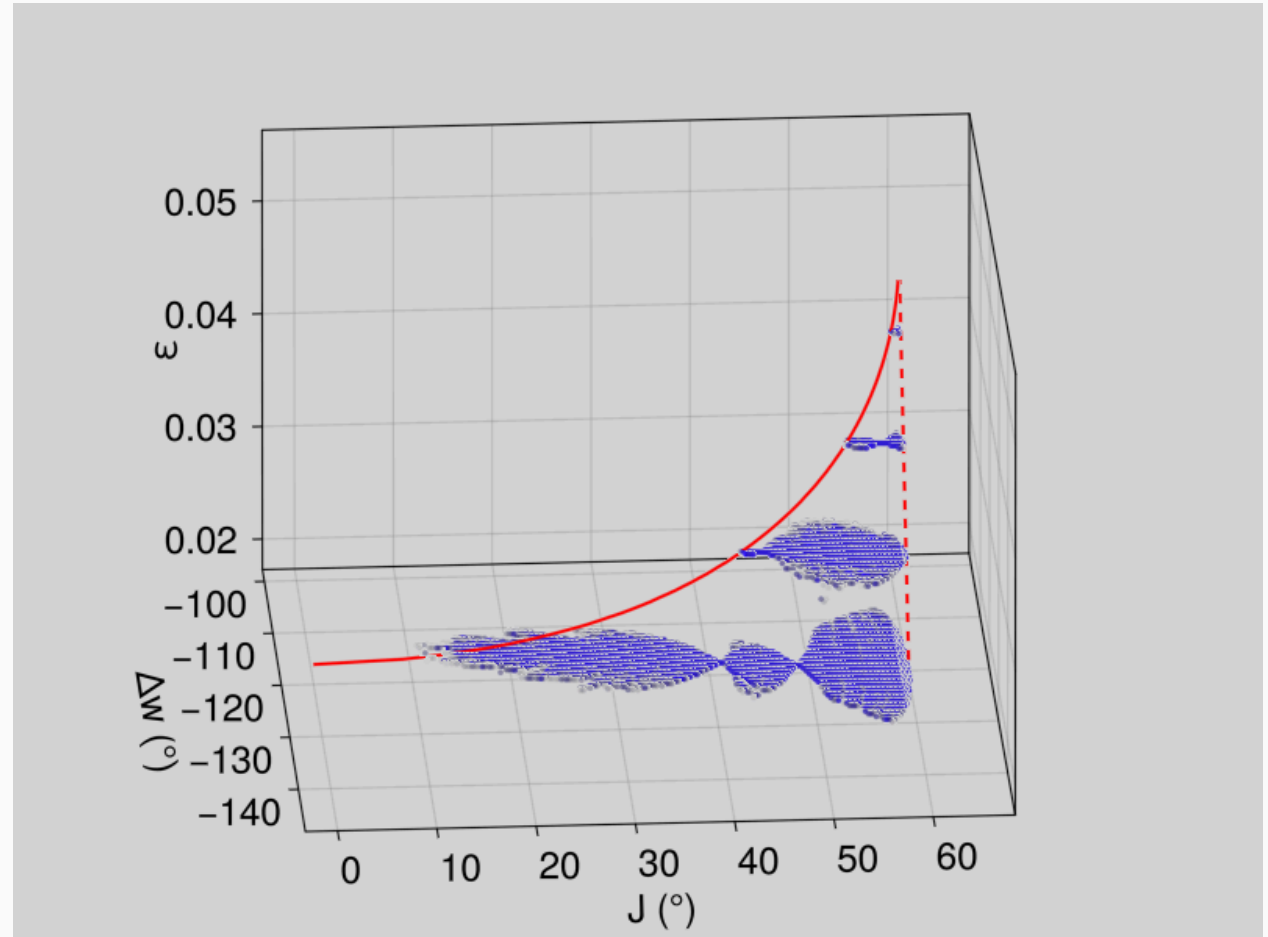
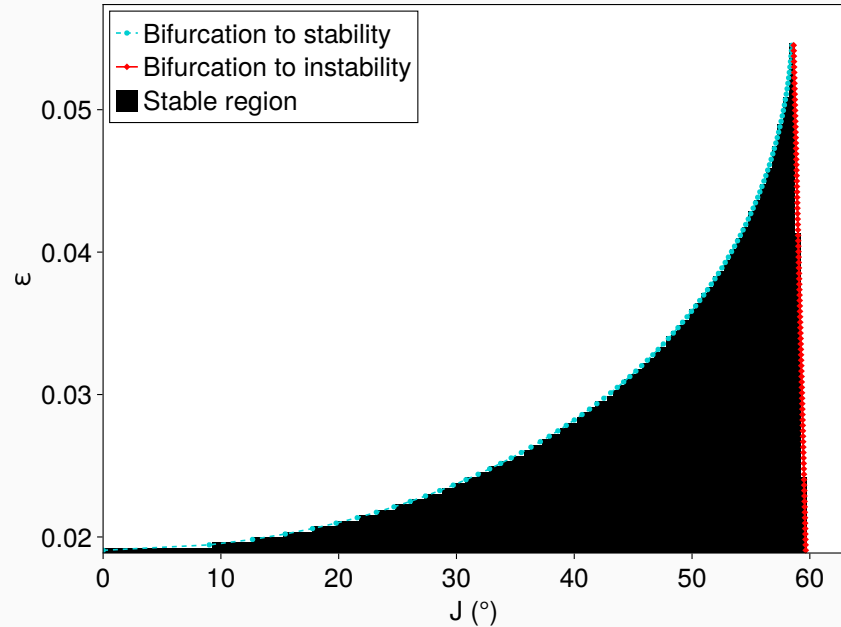
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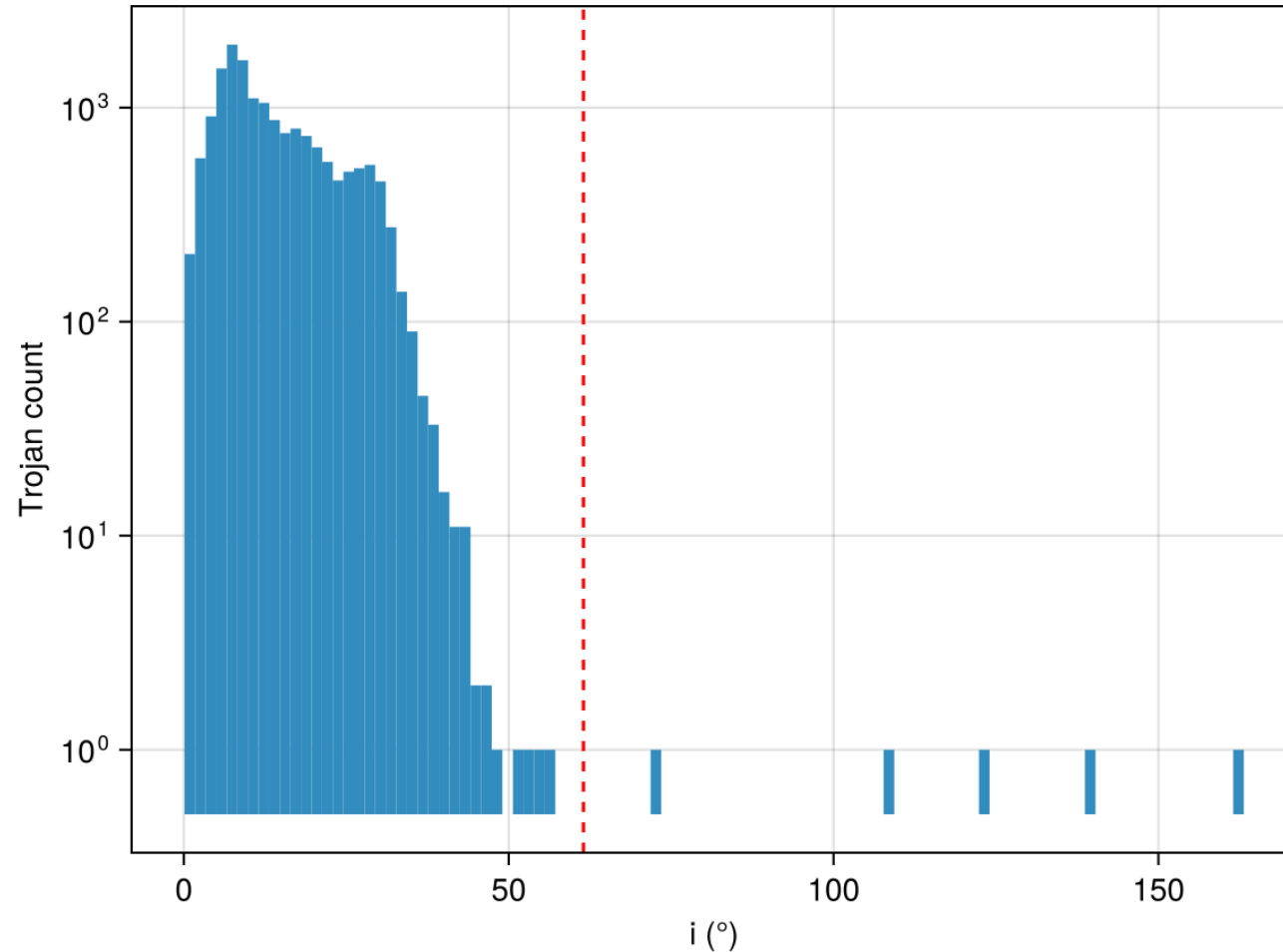
Similar to (Roberts 2002)



# Above Gascheau's value (long-term stability)



# Astronomical consequences? Jupiter's Trojans



**Towards  $P_{12}$**

*Equal masses and bifurcations*

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# Symmetries

## Half-time symmetry

$$\forall \mathbf{x} \in \mathcal{VF}_{\mathcal{L}}, \Phi_{\frac{T}{2}}(\mathbf{x}; C) = -\mathbf{x}$$

$$\Rightarrow f : (\mathbf{x}; C) \mapsto \begin{pmatrix} \Phi_{\frac{T}{2}}(\mathbf{x}; C) + \mathbf{x} \\ \sigma_1(\mathbf{x}; C) \\ \sigma_2(\mathbf{x}; C) \end{pmatrix}$$

## Time-reversal symmetry

$$\forall \mathbf{x} \in \mathcal{VF}_{\mathcal{L}}, t \in \mathbb{R},$$

$$\Phi_{-t}(\mathbf{x}; C) = R_y(\pi) p_{1 \leftrightarrow 2} \Phi_t(\mathbf{x}; C)$$

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Cuts integration time in half

## Time-reversal symmetry

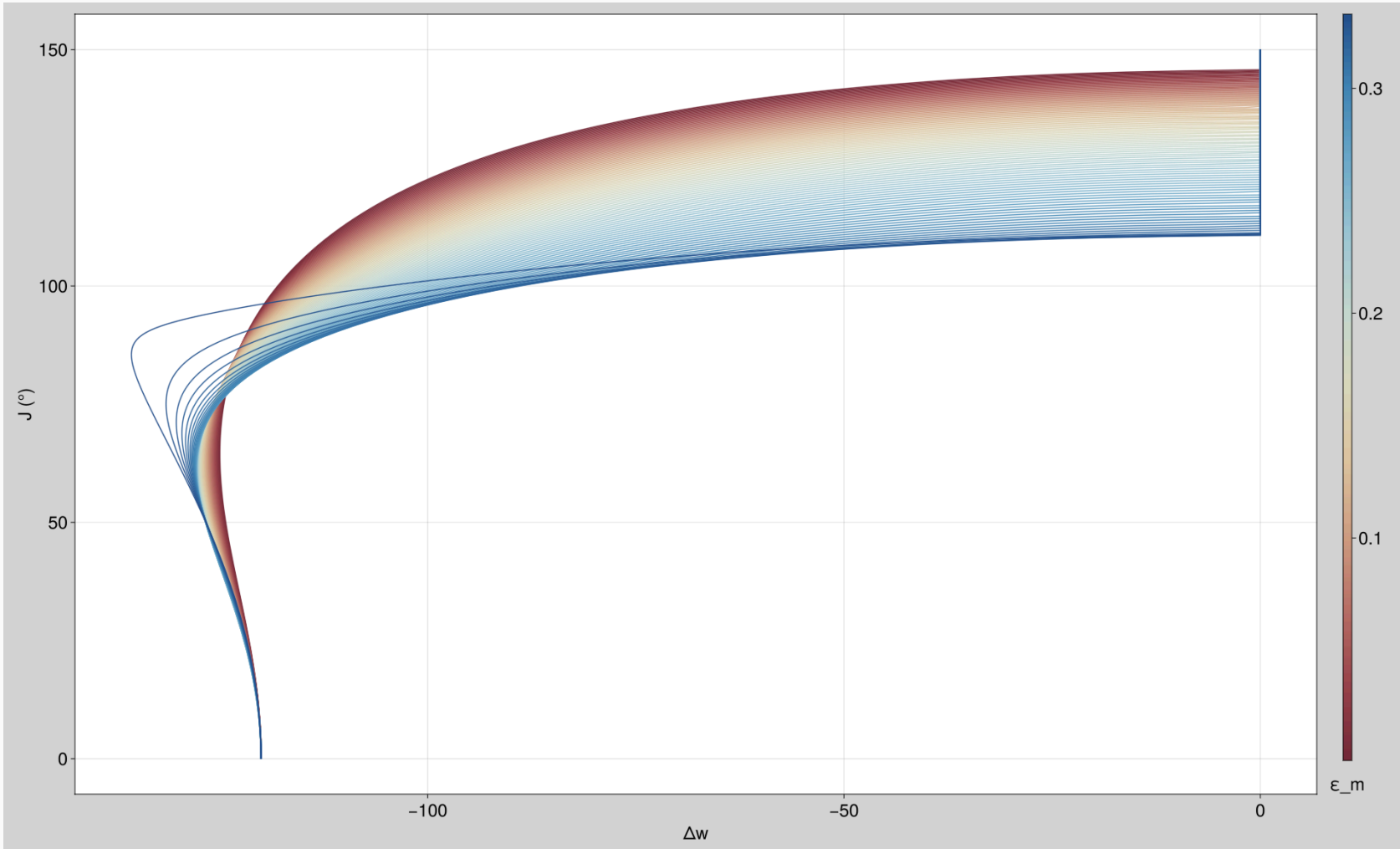
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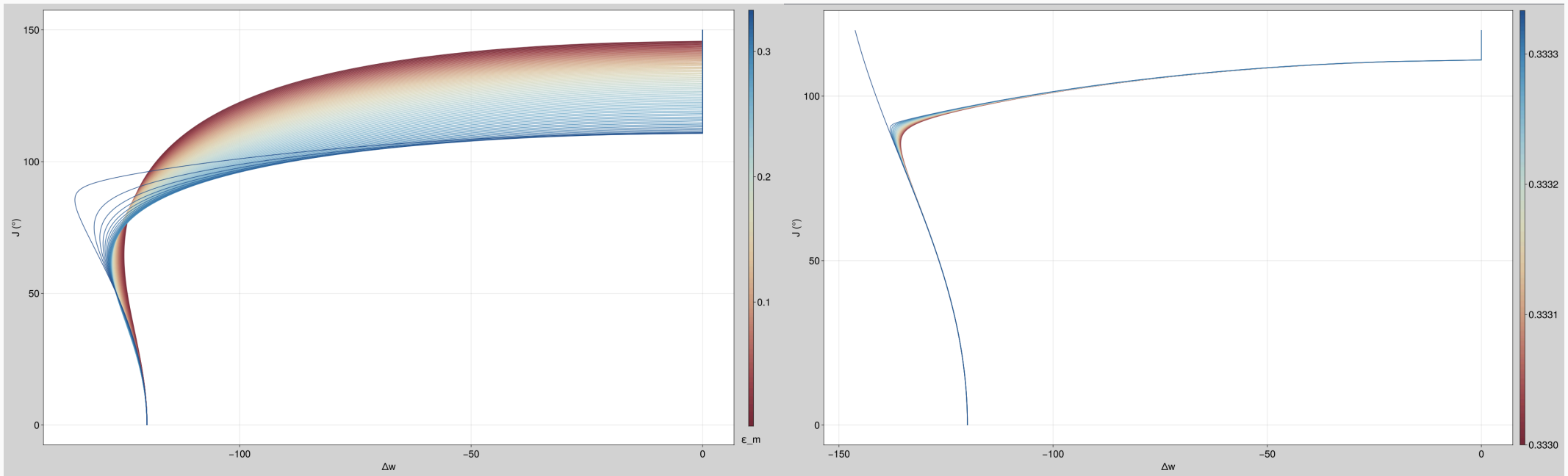
Allows us to add sections

$$\sigma_3(\mathbf{x}; C) = \begin{pmatrix} r_1 - r_2 \\ R_1 + R_2 \\ G_1 - G_2 \end{pmatrix}$$

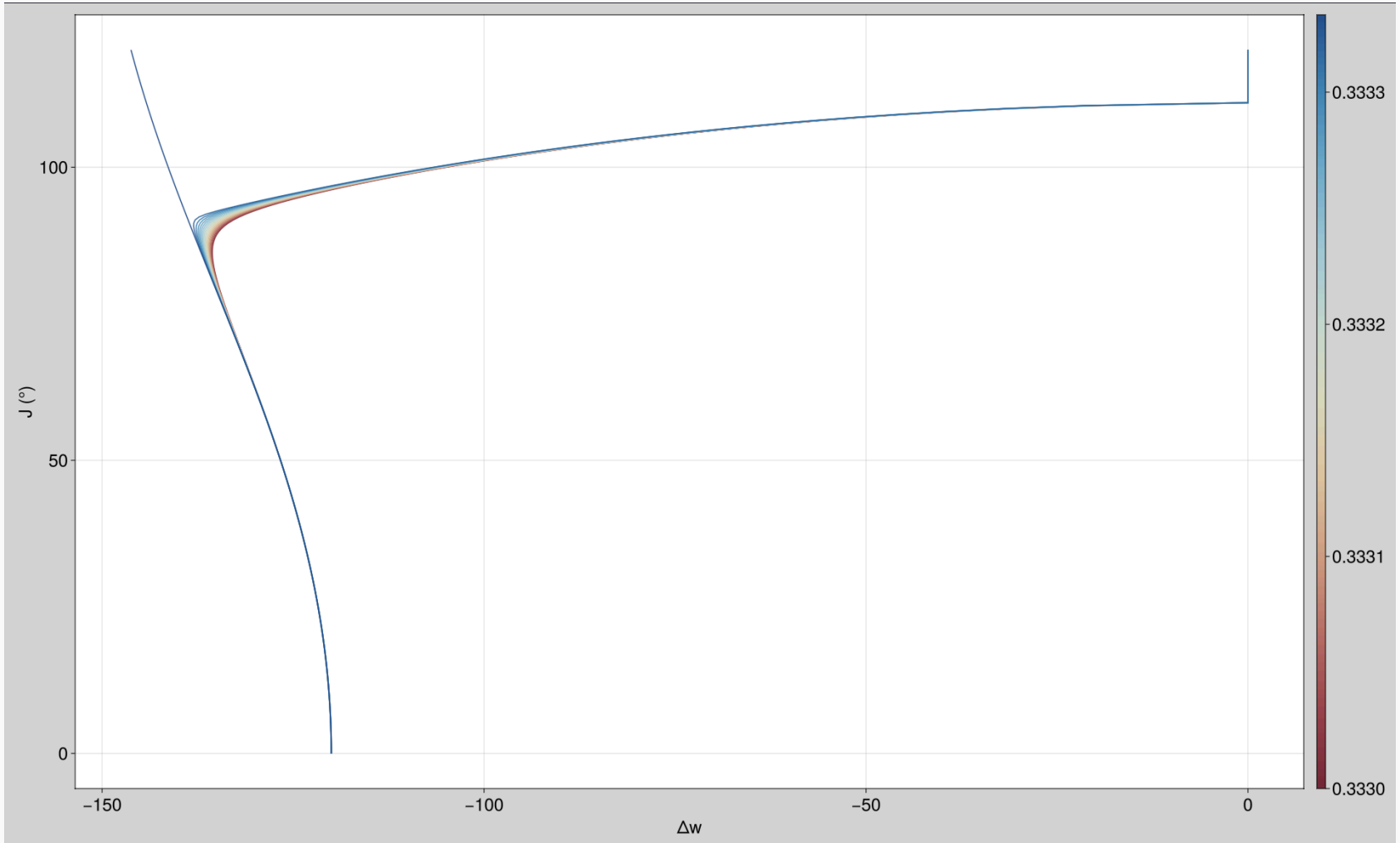
# Approaching equal masses



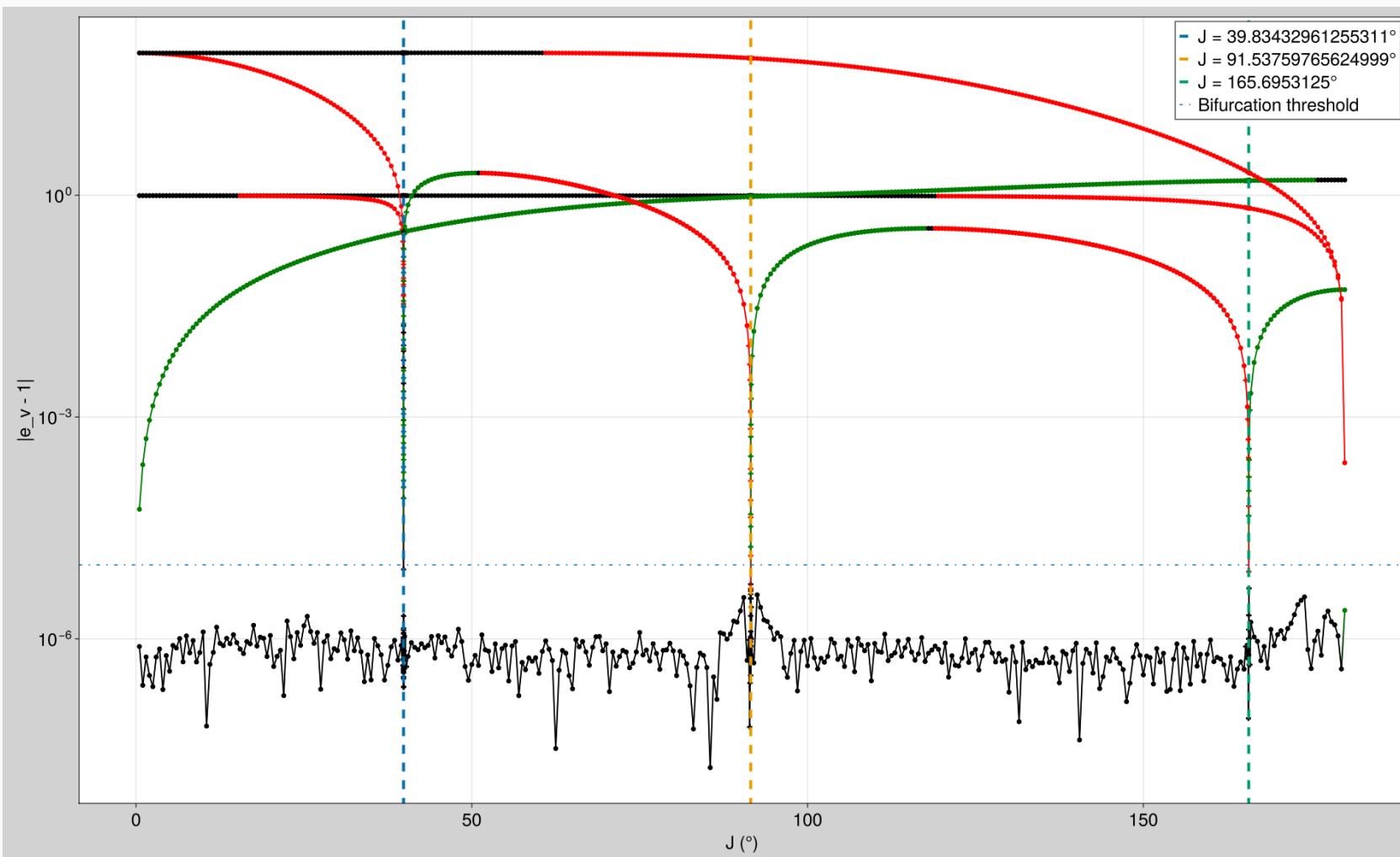
# Approaching equal masses



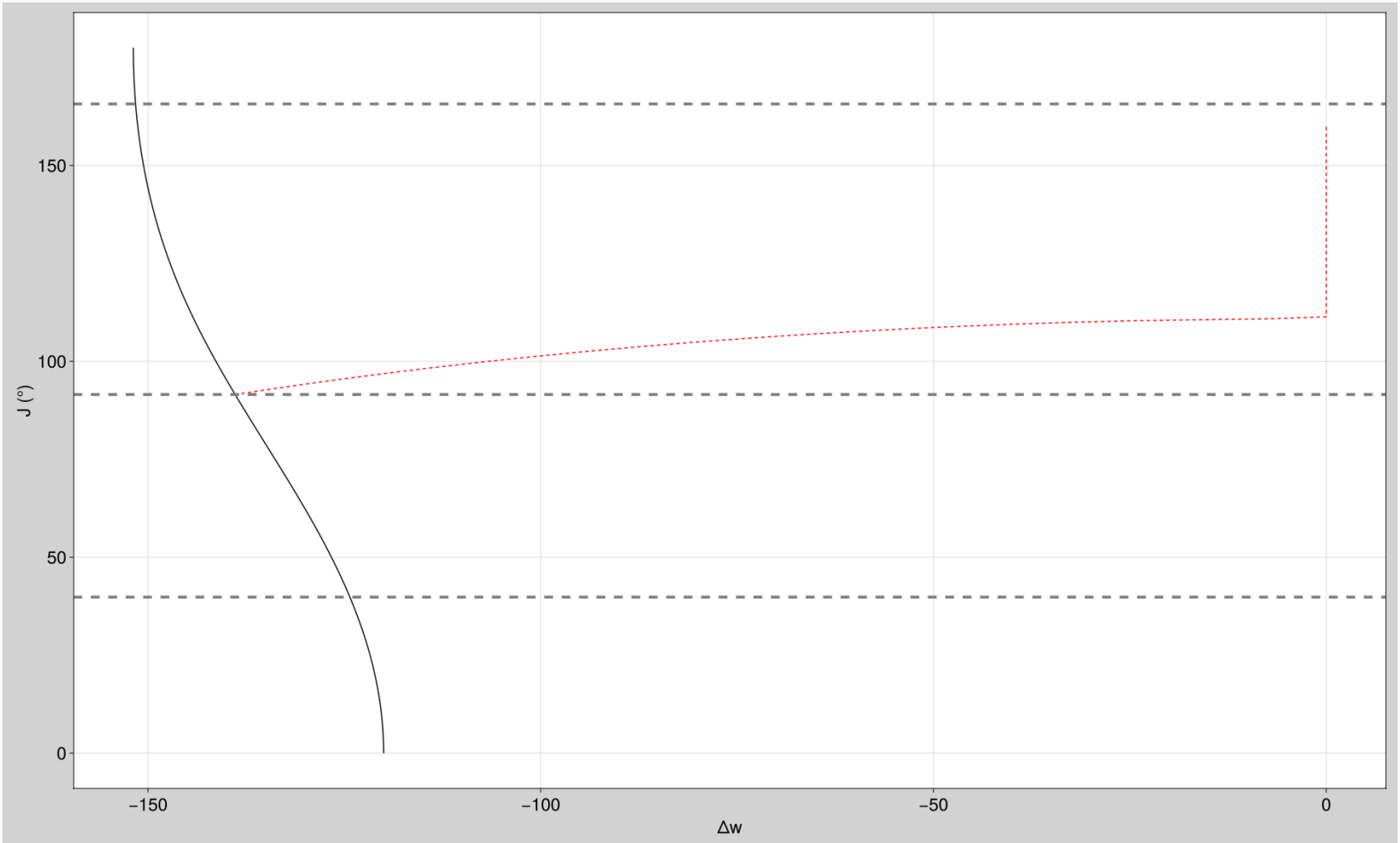
# Approaching equal masses



# Bifurcations at equal masses



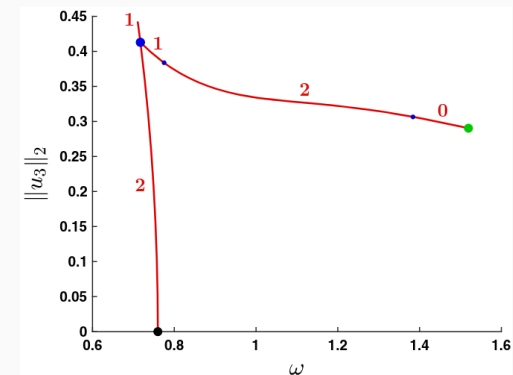
# $\mathcal{VF}_{\mathcal{L}}$ and $P_{12}$ : Marchal's family



# The other bifurcations



- Third bifurcation unexplored for now
- First bifurcation to a “lazy Eight”
  - Cf. (Calleja et al. 2021), and Hénot, Fejoz & Chenciner



# Conclusion

*And further work*

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# Conclusion

We showed...

- Persistence of  $\mathcal{VF}_{\mathcal{L}}$  in the full problem
- Numerical validation of Marchal's generalized conjecture, up to bifurcations

And we developed an open-source numerical toolbox for investigating recurring dynamics

- `TaylorInterface.jl` is available in the Julia General Registry
- Algorithms for planetary results available at [\(Priour 2025\)](#)

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## Publications

- Planetary results at [\(Prieur and Robutel 2026\)](#)
- Bifurcations in preparation

## Further work: bifurcations

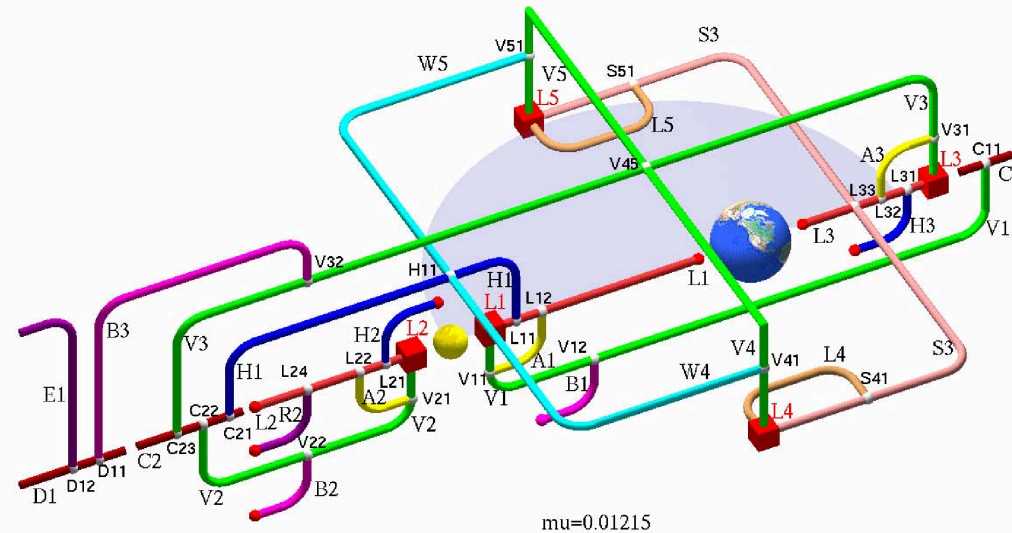


Figure 41: Review of families in the restricted problem. Source: (Doedel et al. 2007, fig 3 and 51).

- Detailed study and following of bifurcations along  $\mathcal{VF}_{\mathcal{L}}$  and  $P_{12}$ 
  - Continue  $P_{12}$  at non-equal masses
- Improve our algorithms for bifurcations (eg. Newton-Raphson)
- Apply `TaylorInterface.jl` and our numerical toolbox to more problems

In memory of  
Christian Marchal  
and  
Àngel Jorba

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